

CNS

UNIT 3

Q1. Explain DSDV (Destination Sequenced Distance Vector) with connection establishment and data transfer phase in detail.

ANS : DSDV (Destination Sequenced Distance Vector)

◆ 1. Introduction

- DSDV is a proactive (table-driven) routing protocol used in Adhoc Wireless Networks.
 - It is based on the Bellman-Ford routing algorithm.
 - Each node maintains a complete routing table for all destinations in the network.
 - It uses sequence numbers generated by destination nodes to ensure loop-free routing.
 - Routes are always available, so no route discovery delay occurs.
-

◆ 2. Routing Table Structure

Each node maintains a routing table with the following fields:

- Destination Address – node to which route is required
- Next Hop – neighbor node to forward packet
- Hop Count – distance to destination
- Sequence Number – indicates freshness of route

👉 Higher sequence number = newer and valid route

 Diagram: Routing Table

Node A Routing Table

Destination | Next Hop | Hop Count | Sequence No

B	B	1	12
C	B	2	14
D	C	3	16

👉 Draw this table neatly in exam ✔

◆ 3. Working of DSDV

- Each node periodically broadcasts routing updates to its neighbors.
- Two types of updates:
 - Full Dump – entire routing table is sent
 - Incremental Update – only changed entries are sent
- When a node receives an update:
 - It compares sequence numbers
 - Selects the route with higher sequence number
 - If same sequence number → choose minimum hop count

◆ 4. Connection Establishment Phase

- DSDV is a proactive protocol, so no separate route discovery process is required.
- All routes are already available in routing table.

Steps:

1. Suppose node A wants to send data to D.
2. Node A checks its routing table.
3. It finds:

- Next hop (e.g., C)
 - Hop count
 - Sequence number
4. If sequence number is latest, route is considered valid.
 5. Connection is established immediately without delay.

👉 Hence, connection setup time is very low.

◆ 5. Data Transfer Phase

Steps:

1. Source node selects the best route from its routing table.
2. Data packet is forwarded to the next hop node.
3. Each intermediate node:
 - Checks its routing table
 - Forwards packet to next hop
4. Process continues until packet reaches destination.

 Diagram: Data Transmission

A ----> B ----> C ---->

Src) (Intermediate Nodes) (Dest)

👉 Draw this diagram for data flow ✓

◆ 6. Role of Sequence Numbers

- Generated by destination nodes
- Used to:
 - Avoid routing loops

- Identify fresh routes
 - Rules:
 - Higher sequence number → better route
 - Even number → valid link
 - Odd number → broken link
-

◆ 7. Advantages

- Simple and easy to implement
 - No delay in route discovery
 - Loop-free routing due to sequence numbers
 - Suitable for small and less dynamic networks
-

◆ 8. Disadvantages

- High control overhead due to frequent updates
 - Wastes bandwidth even when no data is sent
 - Not suitable for highly dynamic networks
 - Routing table maintenance is complex for large networks
-

◆ 9. Conclusion

- DSDV is a proactive routing protocol where routes are always maintained.
- It provides fast communication but suffers from high overhead.
- Best suited for small and stable Adhoc networks.

Q2. Comment on Adhoc Network MAC Layer with Design Issues, Design Goal & Classification.

ANS : Adhoc Network MAC Layer

◆ 1. Introduction

- The MAC (Medium Access Control) layer in Adhoc networks controls how nodes share the wireless medium.
 - It is responsible for channel access, collision avoidance, and fair usage.
 - Unlike wired networks, there is no centralized control in Adhoc networks.
 - Nodes communicate in a distributed manner.
-

◆ 2. Design Issues in Adhoc Network MAC Layer

1. Hidden Terminal Problem

- Two nodes cannot hear each other but send data to the same receiver.
- Causes collision at receiver.

👉 Example: A and C send to B, but A & C cannot hear each other.

2. Exposed Terminal Problem

- A node refrains from sending data even when it is safe.
 - Leads to underutilization of channel.
-

Diagram: Hidden & Exposed Terminal

HiddenTerminal:

A ----> B <---- C (A and C can't hear each other → collision)

ExposedTerminal:

A ----> B C ----> D (C avoids sending unnecessarily)

👉 Draw this diagram in exam ✔

3. Dynamic Topology

- Nodes move frequently → network topology changes rapidly.
 - MAC protocol must adapt quickly.
-

4. Limited Bandwidth

- Wireless medium has low bandwidth compared to wired networks.
 - Efficient utilization is required.
-

5. Energy Constraints

- Nodes are battery-powered.
 - MAC layer must minimize energy consumption.
-

6. Fairness Problem

- All nodes should get equal chance to access channel.
 - Avoid starvation of nodes.
-

7. Collision Avoidance

- Collisions waste bandwidth.
- MAC must use techniques like RTS/CTS to avoid collisions.

◆ 3. Design Goals of MAC Layer

1. Efficient Channel Utilization

- Maximize data transmission using available bandwidth.
 - Reduce idle time and collisions.
-

2. Fairness

- Ensure all nodes get equal opportunity to transmit data.
 - Avoid domination by a single node.
-

3. Low Delay

- Minimize delay in accessing channel.
 - Important for real-time applications.
-

4. Energy Efficiency

- Reduce power consumption.
 - Important for battery-operated nodes.
-

5. Scalability

- Protocol should work for small and large networks.
-

6. Collision Avoidance

- Reduce packet loss due to collisions.
 - Use control packets like RTS/CTS.
-

7. Distributed Operation

- No central controller.
 - Nodes independently decide channel access.
-

◆ 4. Classification of MAC Protocols

1. Contention-Based MAC Protocols

- Nodes compete for channel access.
- Use CSMA/CA mechanism.

Examples:

- IEEE 802.11
- MACAW

Features:

- Simple and flexible
 - May cause collisions
-

2. Contention-Free MAC Protocols

- Channel is divided into time slots.
- Each node gets fixed time to transmit.

Examples:

- TDMA (Time Division Multiple Access)

Features:

- No collision
 - Requires synchronization
-

3. Hybrid MAC Protocols

- Combination of contention-based and contention-free.

- Provides better performance.

Features:

- Flexible and efficient
- Used in advanced networks

Q3. Briefly explain classification of routing protocol for AdHoc Wireless Networks.

ANS : Classification of Routing Protocols in Adhoc Wireless Networks

◆ **1. Introduction**

- In Adhoc Wireless Networks, routing protocols are used to discover and maintain routes between nodes.
 - Due to dynamic topology, node mobility, and limited bandwidth, routing becomes challenging.
 - Based on how routes are created and maintained, routing protocols are classified into three main types:
 - Proactive (Table-driven)
 - Reactive (On-demand)
 - Hybrid
-

◆ **3. Proactive (Table-Driven) Routing Protocols**

◆ **Explanation**

- In proactive protocols, each node maintains routing information to all other nodes in the network.
- Routing tables are updated periodically, even if no data transmission is required.
- Based on distance vector or link state algorithms.

◆ Working

1. Each node maintains a routing table.
 2. Periodically shares updates with neighbors.
 3. Routes are always ready before data transmission.
-

◆ Features

- Immediate route availability
 - No route discovery delay
 - High control overhead due to frequent updates
-

◆ Advantages

- Fast data transmission
 - Suitable for small networks
-

◆ Disadvantages

- Wastes bandwidth due to unnecessary updates
 - Not efficient for highly dynamic networks
-

◆ Examples

- DSDV (Destination Sequenced Distance Vector)
 - OLSR (Optimized Link State Routing)
-

◆ 4. Reactive (On-Demand) Routing Protocols

◆ Explanation

- Routes are created only when required by the source node.
 - Uses route discovery and route maintenance mechanisms.
-

◆ Working

1. Source node broadcasts Route Request (RREQ).
 2. Destination replies with Route Reply (RREP).
 3. Route is established and used for data transfer.
-

◆ Features

- Low control overhead
 - Efficient bandwidth usage
 - Higher delay during route discovery
-

◆ Advantages

- Suitable for large and dynamic networks
 - Reduces unnecessary routing updates
-

◆ Disadvantages

- Initial delay before communication
 - Flooding of RREQ increases overhead
-

◆ Examples

- AODV (Adhoc On-Demand Distance Vector)
- DSR (Dynamic Source Routing)

◆ 5. Hybrid Routing Protocols

◆ Explanation

- Combines features of both proactive and reactive protocols.
- Network is divided into zones or regions.

◆ Working

- Within zone → Proactive routing
- Between zones → Reactive routing

◆ Features

- Reduces routing overhead
- Improves scalability

◆ Advantages

- Balanced performance
- Efficient for large networks

◆ Disadvantages

- Complex to implement
- Zone maintenance required

◆ Examples

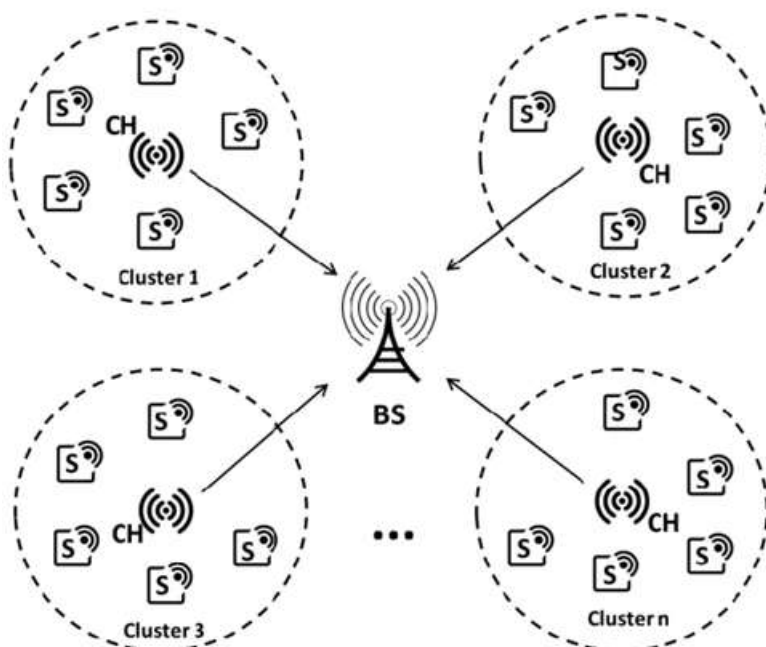
- ZRP (Zone Routing Protocol)

◆ 6. Comparison of Routing Protocols

Parameter	Proactive	Reactive	Hybrid
Route Availability	Always available	On-demand	Partial
Control Overhead	High	Low	Medium
Delay	Low	High	Medium
Scalability	Low	High	High

Q4. Explain with diagram Clustered Architecture of Sensor Network.

ANS :



Clustered Architecture of Sensor Network

◆ 1. Introduction

- In Wireless Sensor Networks (WSN), nodes are organized into clusters to improve efficiency.
 - Each cluster has a Cluster Head (CH) and multiple sensor nodes.
 - Cluster Head manages communication within the cluster and with the base station.
 - This architecture helps in energy saving and scalability.
-

◆ 2. Components of Clustered Architecture

1. Sensor Nodes

- Small devices that sense data from environment (temperature, humidity, etc.).
 - Limited battery, processing, and communication capability.
-

2. Cluster Head (CH)

- Special node with higher energy and processing power.
 - Responsible for:
 - Collecting data from sensor nodes
 - Data aggregation
 - Sending data to base station
-

3. Base Station (Sink)

- Central node that receives data from cluster heads.
- Performs data processing and analysis.
- Acts as interface between network and user.

◆ 3. Working of Clustered Architecture

1. Network is divided into multiple clusters.
2. Each cluster selects a Cluster Head (CH).
3. Sensor nodes send data to their respective CH.
4. CH performs data aggregation (removes redundancy).
5. Aggregated data is transmitted to the base station.

👉 This reduces energy consumption and communication overhead.

◆ 5. Advantages

- Energy efficient due to data aggregation
- Reduces communication overhead
- Improves scalability
- Extends network lifetime

◆ 6. Disadvantages

- Failure of cluster head affects entire cluster
- Cluster head consumes more energy
- Requires cluster formation and management

Q5. Write a short note on MACAW

ANS :

MACAW (Multiple Access with Collision Avoidance for Wireless)

◆ **1. Introduction**

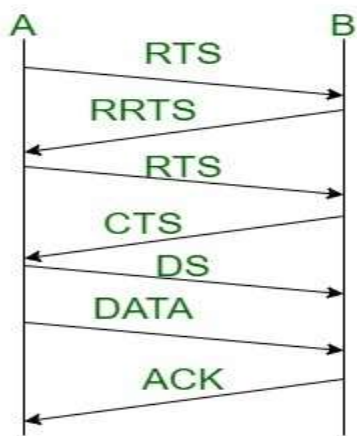
- MACAW is an improved version of the MACA (Multiple Access with Collision Avoidance) protocol.
 - It is designed for Adhoc Wireless Networks to reduce collisions and improve fairness.
 - Developed to overcome limitations of MACA such as unfair bandwidth usage.
-

◆ **2. Working of MACAW**

- MACAW uses a five-step handshake mechanism for data transmission:

Steps:

1. **RTS (Request to Send)** – Sender requests permission to transmit
2. **CTS (Clear to Send)** – Receiver grants permission
3. **DS (Data Sending)** – Sender informs about data transmission
4. **DATA** – Actual data packet is sent
5. **ACK (Acknowledgement)** – Receiver confirms successful reception



3. Features of MACAW

- **Introduces ACK packet for reliability**
 - **Uses DS packet to inform neighbors**
 - **Improves fairness among nodes**
 - **Reduces hidden terminal problem**
-

◆ 4. Advantages

- **Better performance than MACA**
 - **Reduces collisions**
 - **Provides reliable communication**
 - **Ensures fair channel access**
-

◆ 5. Disadvantages

- **Increased control overhead due to extra packets**
- **Slightly higher delay due to multiple steps**

Q6. Classify protocols used in Sensor Network

ANS :

Classification of Protocols in Sensor Networks (WSN)

◆ 1. Introduction

- **Wireless Sensor Networks (WSN) consist of large number of sensor nodes used for monitoring physical conditions.**
- **These nodes have limited energy, memory, and processing power, so efficient protocols are required.**
- **Protocols are designed to optimize energy consumption, data transmission, and scalability.**

- Sensor network protocols are mainly classified based on network structure and routing mechanism.

◆ 3. Flat Routing Protocols (Data-Centric Protocols)

◆ Explanation

- In flat routing, all sensor nodes have equal roles and functionality.
 - There is no cluster or hierarchy in the network.
 - Communication is data-centric, meaning data is more important than node identity.
-

◆ Working

1. Data is requested using queries.
 2. Nodes that have required data respond.
 3. Data is transmitted using flooding or negotiation techniques.
-

◆ Features

- Simple network structure
 - No need for cluster formation
 - Supports data aggregation in some cases
-

◆ Advantages

- Easy to design and implement
 - Suitable for small-scale networks
-

◆ Disadvantages

- High energy consumption due to flooding

- Redundant data transmission
 - Poor scalability for large networks
-

◆ Examples

- SPIN (Sensor Protocol for Information via Negotiation)
 - Directed Diffusion
-

◆ 4. Hierarchical Routing Protocols (Cluster-Based)

◆ Explanation

- Network is divided into clusters for efficient management.
 - Each cluster has a Cluster Head (CH) and multiple sensor nodes.
 - CH performs data aggregation and communication with base station.
-

◆ Working

1. Clusters are formed in the network.
 2. A node is selected as Cluster Head (CH).
 3. Sensor nodes send data to CH.
 4. CH aggregates data and sends it to Base Station.
-

◆ Features

- Reduces energy consumption
- Supports data aggregation
- Improves scalability

◆ Advantages

- Energy efficient
 - Extends network lifetime
 - Reduces communication overhead
-

◆ Disadvantages

- CH failure affects entire cluster
 - Additional overhead for cluster formation
-

◆ Examples

- LEACH (Low-Energy Adaptive Clustering Hierarchy)
 - PEGASIS
-

◆ 5. Location-Based Routing Protocols

◆ Explanation

- Routing decisions are based on geographical location of nodes.
 - Nodes use GPS or position information to forward data.
-

◆ Working

1. Each node knows its location coordinates.
2. Data is forwarded to nodes closer to destination.
3. Reduces unnecessary transmissions.

◆ Features

- Efficient routing using location awareness
 - Reduces energy consumption
-

◆ Advantages

- Scalable for large networks
 - Minimizes communication cost
-

◆ Disadvantages

- Requires GPS or location tracking system
 - Increases cost and complexity
-

◆ Examples

- GAF (Geographic Adaptive Fidelity)
 - GPSR (Greedy Perimeter Stateless Routing)
-

◆ 6. Other Important Classifications (Extra Points)

1. Energy-Aware Protocols

- Designed to minimize energy consumption.
 - Select routes based on energy availability.
-

2. Multipath Routing Protocols

- Uses multiple paths for data transmission.
 - Improves reliability and fault tolerance.
-

3. Query-Based Protocols

- Data transmission based on queries from base station.
-

7. Comparison (Short Table)

Parameter	Flat Routing	Hierarchical Routing	Location-Based Routing
Structure	No hierarchy	Cluster-based	Location-based
Energy Efficiency	Low	High	High
Scalability	Low	High	High
Complexity	Low	Medium	Medium

Q7. Explain AODV and DSR protocol in detail.

ANS :

AODV and DSR Routing Protocols in Adhoc Networks

◆ 1. Introduction

- AODV (Adhoc On-Demand Distance Vector) and DSR (Dynamic Source Routing) are reactive (on-demand) routing protocols.
 - Routes are created only when required, reducing unnecessary overhead.
 - Both are widely used in Adhoc Wireless Networks.
-

◆ 2. AODV (Adhoc On-Demand Distance Vector)

◆ Explanation

- AODV is a reactive routing protocol that establishes routes only when needed.
- It uses routing tables and sequence numbers to maintain fresh routes.

◆ Working of AODV

1. Route Discovery

1. Source node sends RREQ (Route Request) message.
2. RREQ is broadcast to all neighbors.
3. Intermediate nodes forward RREQ further.
4. Destination or intermediate node with fresh route replies with RREP (Route Reply).
5. Route is established

A ~~~> B ~~~> C ~~~> D

(S) () () (D)

RREQ broadcast from A to D

RREP sent back from D to A

2. Data Transfer

- After route establishment, data packets are sent through selected path.
- Each node maintains next hop information in routing table.

3. Route Maintenance

- If link breaks:
 - Route Error (RERR) message is sent
 - Route is deleted
 - New route discovery is initiated

◆ Features

- Uses sequence numbers to avoid loops
- Maintains routing tables
- Supports dynamic topology

◆ Advantages

- Less overhead than proactive protocols
- Loop-free routing
- Suitable for large networks

◆ Disadvantages

- Initial delay due to route discovery
- Flooding of RREQ increases overhead

◆ 3. DSR (Dynamic Source Routing)

◆ Explanation

- DSR is a reactive routing protocol based on source routing.
- Entire route is stored in packet header.

◆ Working of DSR

1. Route Discovery

1. Source node broadcasts RREQ.
2. Each node appends its address to RREQ.
3. When RREQ reaches destination, it sends RREP with complete route.

Diagram: DSR Route Discovery

A → B → C → D

route stored in packet:

A-B-C-D

2. Data Transfer

- Source node sends data with complete route in packet header.
 - Intermediate nodes simply forward packet based on given route.
-

3. Route Maintenance

- If link fails:
 - Error message is sent to source
 - Source uses alternate route from cache or starts new discovery
-

◆ Features

- Uses source routing
 - Maintains route cache
 - No need for periodic updates
-

◆ Advantages

- No routing table required at intermediate nodes
 - Efficient for small networks
 - Multiple routes stored in cache
-

◆ **Disadvantages**

- Packet header size increases (due to full path)
- Not suitable for large networks
- High overhead in long routes

◆ **4. Difference Between AODV and DSR**

Parameter	AODV	DSR
Routing Type	Table-driven (on-demand)	Source routing
Route Storage	Routing table	Packet header
Sequence Number	Used	Not used
Overhead	Moderate	High (header overhead)
Scalability	Better for large networks	Suitable for small networks

Q8. Explain the issues in designing a Routing Protocol for Ad-hoc Wireless Networks

ANS :

Issues in Designing a Routing Protocol for Adhoc Wireless Networks

◆ **1. Introduction**

- Adhoc Wireless Networks have no fixed infrastructure and nodes communicate dynamically.
- Designing a routing protocol is difficult due to mobility, limited resources, and dynamic topology.
- A good routing protocol must ensure reliable and efficient communication.

◆ 2. Issues in Designing Routing Protocol

◆ 1. Dynamic Topology

- Nodes move frequently, causing frequent changes in network structure.
- Routes may break often, requiring continuous updates.

◆ 2. Limited Bandwidth

- Wireless links have low bandwidth compared to wired networks.
- Routing protocols must minimize control overhead.

◆ 3. Energy Constraints

- Nodes are battery-powered with limited energy supply.
- Protocol must reduce energy consumption to increase network lifetime.

◆ 4. Routing Overhead

- Frequent routing updates increase network traffic.
- Excess overhead reduces overall performance.

◆ 5. Scalability

- Network size may increase dynamically.
- Protocol should perform efficiently for small and large networks.

◆ 6. Quality of Service (QoS)

- Some applications require low delay, high reliability, and bandwidth guarantee.

- Providing QoS in Adhoc networks is difficult.
-

◆ 7. Security Issues

- Open wireless medium is vulnerable to attacks (e.g., eavesdropping, spoofing).
 - Protocol must ensure secure routing.
-

◆ 8. Hidden and Exposed Terminal Problems

- Causes packet collisions or underutilization of channel.
 - Affects routing performance.
-

◆ 9. Unidirectional Links

- Sometimes communication is possible only in one direction.
 - Routing becomes complex in such cases.
-

◆ 10. Mobility-Induced Route Breakage

- High mobility leads to frequent link failures.
 - Protocol must quickly detect and repair routes.
-

◆ 11. Load Balancing

- Some nodes may be overused, causing energy depletion.
 - Routing should distribute traffic evenly.
-

◆ 12. Delay Constraints

- Route discovery and maintenance introduce delay.

- Important for real-time applications.

Q9. Explain design issues and design goal in adhoc network MAC layer.

ANS:

Design Issues and Design Goals in Adhoc Network MAC Layer

◆ 1. Introduction

- The MAC (Medium Access Control) layer controls how nodes access the shared wireless medium.
 - In Adhoc networks, there is no centralized control, so MAC must work in a distributed manner.
 - It plays an important role in collision avoidance, fairness, and efficient communication.
-

◆ 2. Design Issues in Adhoc Network MAC Layer

◆ 1. Hidden Terminal Problem

- Two nodes cannot hear each other but transmit to the same receiver.
 - Causes packet collision at receiver.
-

◆ 2. Exposed Terminal Problem

- A node unnecessarily stops transmission even when channel is free.
 - Leads to poor channel utilization.
-

 **Diagram: Hidden & Exposed Terminal**

Hidden Terminal:

A ----> B <---- C

Exposed Terminal:

A ----> B C ----> D

◆ **3. Dynamic Topology**

- **Nodes move frequently, causing frequent changes in network structure.**
 - **MAC protocol must adapt quickly.**
-

◆ **4. Limited Bandwidth**

- **Wireless medium has limited capacity.**
 - **Efficient usage is necessary.**
-

◆ **5. Energy Constraints**

- **Nodes operate on battery power.**
 - **MAC must reduce energy consumption.**
-

◆ **6. Collision Avoidance**

- **Collisions waste bandwidth and energy.**
 - **MAC must use techniques like RTS/CTS.**
-

◆ **7. Fairness**

- **All nodes should get equal chance to access channel.**
 - **Avoid starvation.**
-

◆ **8. Synchronization**

- **Some protocols require time synchronization among nodes.**

- **Difficult in distributed environment.**
-

◆ 3. Design Goals of MAC Layer

◆ 1. Efficient Channel Utilization

- **Maximize data transmission using available bandwidth.**
 - **Minimize idle time and collisions.**
-

◆ 2. Fairness

- **Ensure equal access to all nodes.**
 - **Prevent any node from dominating the channel.**
-

◆ 3. Low Delay

- **Reduce time required to access channel.**
 - **Important for real-time applications.**
-

◆ 4. Energy Efficiency

- **Minimize power consumption of nodes.**
 - **Extend network lifetime.**
-

◆ 5. Scalability

- **Protocol should work efficiently for large number of nodes.**
-

◆ 6. Collision Avoidance

- **Reduce packet loss using mechanisms like RTS/CTS.**

◆ **7. Distributed Operation**

- No central controller.
 - Nodes independently decide when to transmit.
-

◆ **8. Throughput Maximization**

- Increase successful data transmission rate.
- Reduce retransmissions.

Q10 Compare sensor network with adhoc wireless network.

ANS:

Parameter	Adhoc Wireless Network	Sensor Network (WSN)
Purpose	General communication between nodes	Monitoring and sensing environment
Node Type	Mobile devices (laptops, mobiles)	Sensor nodes with sensing capability
Node Mobility	High mobility	Mostly static or low mobility
Energy Constraint	Moderate	Very high (battery limited)
Data Transmission	Point-to-point communication	Data-centric (many-to-one)
Network Size	Small to medium	Large-scale (hundreds/thousands nodes)
Processing Power	Higher	Very limited
Communication Pattern	Any node to any node	Sensor nodes → Base station

Parameter	Adhoc Wireless Network	Sensor Network (WSN)
Routing Protocols	AODV, DSR, DSDV	LEACH, PEGASIS, SPIN
Cost of Nodes	Expensive	Low-cost nodes
Deployment	Planned or random	Mostly random deployment
Fault Tolerance	Moderate	High (due to large number of nodes)

Q11. Explain DSR in details with route discovery & route maintenance.

ANS:

DSR (Dynamic Source Routing) Protocol

◆ 1. Introduction

- DSR is a reactive (on-demand) routing protocol used in Adhoc Wireless Networks.
- Routes are created only when required, reducing overhead.
- It uses source routing, where the complete path is stored in the packet header.
- No need for periodic routing table updates.

◆ 2. Key Features of DSR

- Uses Route Discovery and Route Maintenance
 - Maintains Route Cache (stores multiple routes)
 - No routing tables at intermediate nodes
 - Supports dynamic topology
-

◆ 3. Route Discovery in DSR

◆ Steps:

1. Source node (A) wants to send data to destination (D).
 2. A broadcasts RREQ (Route Request) packet.
 3. Each intermediate node:
 - Appends its address to RREQ
 - Forwards it further
 4. When RREQ reaches destination (D):
 - D sends RREP (Route Reply) back to source
 5. RREP contains the complete route.
-

Diagram: Route Discovery

A → B → C → D

RREQ Path: A-B-C-D

RREP Path: D-C-B-A

👉 Draw this diagram in exam ✓

◆ 4. Data Transfer Phase

- Source node sends data packets with complete route in header.
- Example: A → B → C → D
- Intermediate nodes simply forward packet as per route.

◆ 5. Route Maintenance in DSR

◆ Steps:

1. If any link breaks (e.g., B–C fails):
 2. Node detecting failure sends Route Error (RERR) message to source.
 3. Source node:
 - Removes broken route from cache
 - Uses alternate route (if available)
 - Otherwise starts new route discovery
-

Diagram: Route Maintenance

A → B X C → D

(LinkFailure)

RERR sent back to A

New route discovery starts

◆ 6. Advantages

- No periodic updates → low overhead
- Multiple routes stored in cache
- Efficient for small networks

◆ 7. Disadvantages

- Packet header size increases (full path stored)
- Not suitable for large networks
- High overhead in long routes

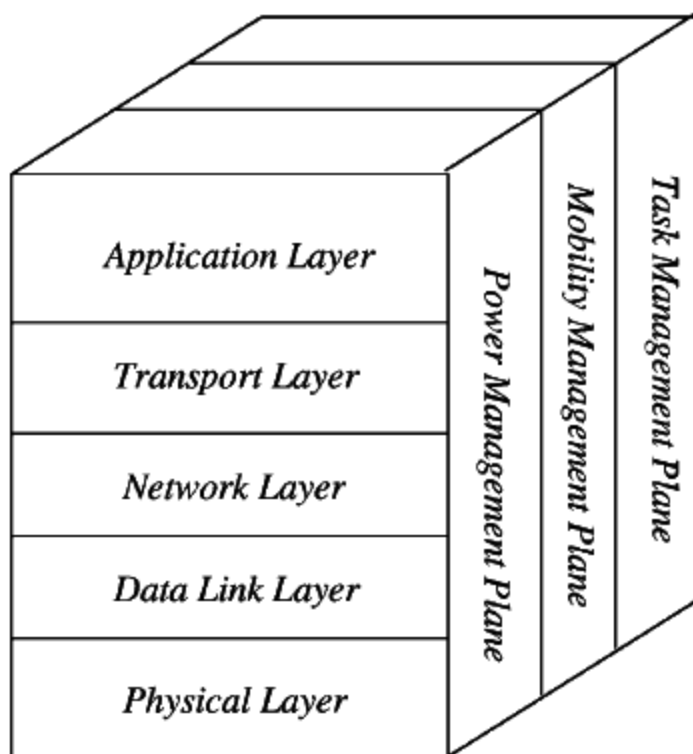
Q12. Explain with diagram layered Architecture of Sensor Network.

ANS:

Layered Architecture of Sensor Network

◆ 1. Introduction

- **Wireless Sensor Network (WSN)** follows a layered architecture similar to computer networks.
- It divides network functions into different layers, each with specific responsibilities.
- This architecture helps in efficient communication, scalability, and energy management.



1. Physical Layer

- Responsible for **signal transmission and reception**.
- Handles **frequency selection, modulation, and data rate**.

- Ensures reliable communication over wireless medium.
-

◆ 2. Data Link Layer

- Responsible for **error detection and correction**.
 - Manages **medium access control (MAC)**.
 - Avoids collisions and ensures **reliable link**.
-

◆ 3. Network Layer

- Handles **routing of data packets**.
 - Finds the best path from sensor nodes to base station.
 - Supports **data aggregation and energy-efficient routing**.
-

◆ 4. Transport Layer

- Ensures **end-to-end data delivery**.
 - Provides **reliability and congestion control**.
 - Maintains communication between sensor nodes and sink.
-

◆ 5. Application Layer

- Provides **user interface and application-specific services**.
 - Used in applications like **environment monitoring, health care, military**.
-

◆ 4. Cross-Layer Functions

◆ 1. Power Management Plane

- Monitors and controls **energy consumption**.
- Extends network lifetime.

♦ **2. Mobility Management Plane**

- Tracks movement of nodes.
 - Maintains network connectivity.
-

♦ **3. Task Management Plane**

- Assigns tasks to sensor nodes.
 - Ensures efficient use of resources.
-

♦ **5. Advantages**

- Organized and modular design
- Efficient communication
- Easy maintenance and scalability
- Energy-efficient operation

Q13. Differentiate between infrastructure network and infrastructure less network.

ANS:

No.	Infrastructure Network	Infrastructure-less Network
1	Uses fixed infrastructure (base station, router)	No fixed infrastructure
2	Centralized control	Distributed control
3	Communication via access point/base station	Direct node-to-node communication

No.	Infrastructure Network	Infrastructure-less Network
4	Pre-planned setup required	Can be set up anytime, anywhere
5	Low mobility support	Supports high mobility
6	More stable and reliable	Less stable due to dynamic nature
7	Higher cost (infrastructure needed)	Low cost (no infrastructure)
8	Easy network management	Difficult to manage
9	Used in Wi-Fi, cellular networks	Used in Adhoc networks, MANET
10	Failure of base station affects network	No single point of failure

Q14. What are hidden station and exposed station problem in WLAN.

ANS:

Hidden Station and Exposed Station Problem in WLAN

◆ 1. Introduction

- In Wireless LAN (WLAN), communication occurs over a shared wireless medium.
- Unlike wired networks, nodes cannot always detect all transmissions due to limited transmission range and signal attenuation.
- This leads to two important problems:
 - Hidden Station Problem
 - Exposed Station Problem

- These problems affect collision control, bandwidth utilization, and overall network performance.
-

◆ 2. Hidden Station Problem

◆ Deep Definition

- A Hidden Station Problem occurs when two or more nodes are outside each other's sensing range, but both are within the range of a common receiver.
 - Since they cannot detect each other's transmission, they assume the channel is free and transmit simultaneously, causing collision at the receiver.
-

◆ Detailed Explanation

- Consider nodes A, B, and C:
 - $A \rightarrow B$ and $C \rightarrow B$
 - A and C are not in each other's range
- When A transmits, C cannot sense it $\rightarrow$ C also transmits
- At node B, signals from A and C overlap, causing collision

👉 This happens because carrier sensing fails in wireless networks.

◆ Effects

- Packet collision at receiver
- Increased retransmissions
- Reduced throughput
- Wastage of bandwidth and energy

◆ Solution: RTS/CTS Mechanism

- A sends RTS to B
- B replies with CTS
- C hears CTS → understands channel is busy → stops transmitting
- Prevents collision

◆ 3. Exposed Station Problem

◆ Deep Definition

- An Exposed Station Problem occurs when a node incorrectly assumes the channel is busy and refrains from transmitting, even though its transmission would not cause interference at the intended receiver.

◆ Detailed Explanation

- Consider nodes A, B, C, and D:
 - A → B
 - C → D
- C hears A's transmission and assumes channel is busy
- But C's transmission to D would not interfere with B
- Still, C remains silent unnecessarily

👉 This leads to underutilization of the channel.

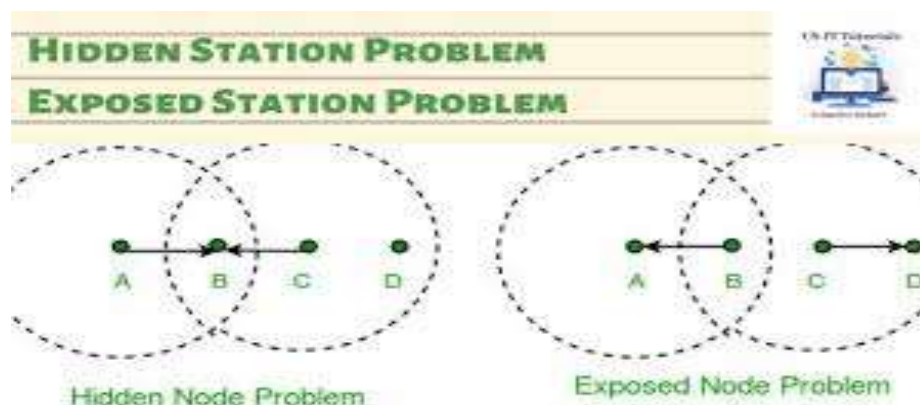
◆ Effects

- Reduced channel utilization

- Increased delay
- Lower network efficiency

◆ Solution

- RTS/CTS helps node decide correctly:
 - If C does not receive CTS from B, it can transmit safely
- Improves channel usage



Q15.Explain Distributed Denial of Service attacks.

ANS:

Distributed Denial of Service (DDoS) Attack

◆ 1. Introduction

- A Distributed Denial of Service (DDoS) attack is a type of cyber attack where multiple systems are used to flood a target server or network with traffic.
- The goal is to make the service unavailable to legitimate users.
- It is called “distributed” because the attack comes from many different sources.

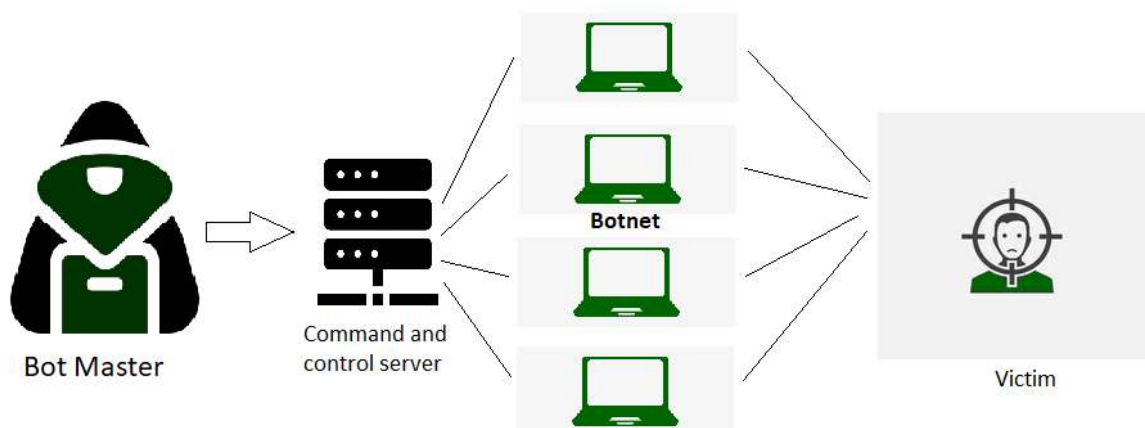
◆ 2. Definition

- DDoS attack is an attack in which multiple compromised computers (bots) send a large number of requests to a target system, causing it to slow down or crash.

◆ 3. Working of DDoS Attack

Steps:

1. Attacker infects many computers with malware → creates a botnet.
2. These infected devices (bots) are controlled remotely.
3. Attacker sends command to all bots.
4. Bots send huge traffic to the target server.
5. Server becomes overloaded and stops responding to real users.



Prevention Techniques

◆ 1. Firewalls and IDS/IPS

- Detect and block malicious traffic.

◆ 2. Traffic Filtering

- Filters abnormal or fake requests.

◆ 3. Rate Limiting

- Limits number of requests per user/IP.

◆ 4. Load Balancing

- Distributes traffic across multiple servers.

◆ 5. Anti-DDoS Tools

- Cloud-based protection services (e.g., CDN-based filtering).

◆ 8. Advantages (Attacker perspective – theoretical)

- Easy to launch using botnets
- Difficult to trace real attacker
- Highly effective in disrupting services

◆ 9. Disadvantages (Impact on victim)

- Service downtime
- Loss of data availability
- Business disruption
- High recovery cost

Q16. List types of Wireless Sensor Network architecture. Explain the phases of LEACH routing protocol in detail.

ANS:

1. Types of Wireless Sensor Network (WSN) Architecture

Wireless Sensor Networks can be classified based on their structure as:

◆ 1. Flat (Data-Centric) Architecture

- All sensor nodes are equal (no hierarchy).
- Data is collected and sent directly using flooding or query-based methods.
- Example protocols: SPIN, Directed Diffusion

◆ 2. Hierarchical (Clustered) Architecture

- Nodes are divided into clusters.
 - Each cluster has a Cluster Head (CH).
 - CH collects and aggregates data from nodes and sends to base station.
 - Example: LEACH, PEGASIS
-

◆ 3. Location-Based Architecture

- Uses node position (GPS/coordinates) for routing.
 - Data is forwarded based on geographical distance.
 - Example: GPSR, GAF
-

2. LEACH Routing Protocol (Low Energy Adaptive Clustering Hierarchy)

◆ 1. Introduction

- LEACH is a hierarchical clustering-based routing protocol.
- It is designed to reduce energy consumption in WSN.

- Nodes randomly become Cluster Heads (CH) to balance energy usage.
-

◆ 2. Phases of LEACH Protocol

LEACH works in two main phases:

◆ (A) Setup Phase

Steps:

1. Cluster Head Selection

- Each node generates a random number.
- If number is less than threshold → node becomes Cluster Head (CH).

2. Advertisement Phase

- CH broadcasts message: "I am Cluster Head".

3. Cluster Formation

- Normal nodes choose nearest CH based on signal strength.
- Nodes send join request to CH.

4. Schedule Creation

- CH creates a TDMA schedule for members.
 - Each node gets time slot for transmission.
-

Diagram: Setup Phase

Nodes → CH Selection → Cluster Formation → TDMA Schedule

◆ (B) Steady-State Phase

Steps:

1. Sensor nodes send data in allocated time slots.
 2. CH collects data from all nodes.
 3. CH performs data aggregation (removes redundancy).
 4. CH sends compressed data to Base Station.
-

Diagram: Steady-State Phase

Sensor Nodes → Cluster Head → Base Station
(Data Aggregation)

◆ 3. Key Features of LEACH

- Random rotation of CH to save energy
- Data aggregation reduces communication load
- Uses TDMA scheduling
- Fully distributed protocol

◆ 4. Advantages

- Energy efficient
- Extends network lifetime
- Reduces communication overhead
- Scalable for large networks

◆ 5. Disadvantages

- Cluster formation overhead
- CH may die faster due to heavy load
- Not suitable for mobile sensor networks

UNIT 4

Q1. What is Cipher Block Chaining (CBC)? Explain the process and Characteristics of CBC with suitable diagram

ANS:

Cipher Block Chaining (CBC) Mode

◆ 1. Introduction

- CBC (Cipher Block Chaining) is a block cipher encryption mode used in cryptography.
 - In CBC, each plaintext block is XORed with the previous ciphertext block before encryption.
 - It ensures that same plaintext blocks produce different ciphertexts, improving security.
-

◆ 2. Definition

- CBC mode is a block cipher technique where each plaintext block is combined with the previous ciphertext block using XOR operation before encryption.
 - The first block uses an Initialization Vector (IV).
-

◆ 3. Process of CBC Mode

◆ Step-by-step Working:

Step 1: Initialization

- Take Initialization Vector (IV) for first block.

Step 2: First Block Encryption

- Plaintext block P1 is XORed with IV:

- $P_1 \oplus IV = \text{input to encryption}$
 - Result is ciphertext C1.
-

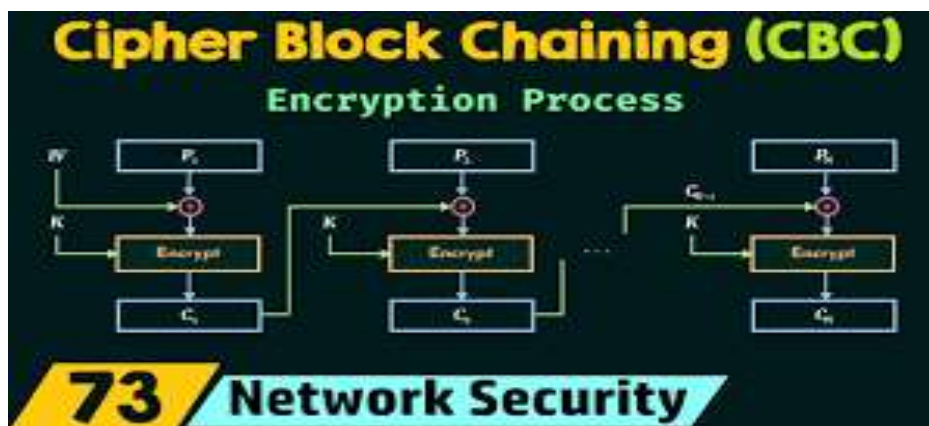
Step 3: Next Blocks

For each block:

- $P_i \oplus C_{i-1} = \text{input to encryption}$
 - Output gives C_i
-

Step 4: Repetition

- Process continues for all plaintext blocks.



5. Decryption Process

- Reverse operation is used:
 - C_i is decrypted first
 - Then XORed with previous ciphertext block
 - Formula:
 - $P_i = \text{Decrypt}(C_i) \oplus C_{i-1}$
-

◆ 6. Characteristics of CBC Mode

- ◆ 1. Uses Initialization Vector (IV)

- First block uses random IV to ensure uniqueness.
-

◆ 2. Chaining Mechanism

- Each ciphertext block depends on previous block.
 - Creates a chain of encryption.
-

◆ 3. Strong Security

- Same plaintext produces different ciphertext each time.
 - Prevents pattern recognition.
-

◆ 4. Error Propagation

- If one ciphertext block is corrupted:
 - Current block gets corrupted
 - Next block is also affected
-

◆ 5. Sequential Processing

- Blocks are processed one after another, not parallel.
 - Slower compared to ECB mode.
-

◆ 6. No Plaintext Pattern Leakage

- Hides repeated patterns in plaintext.
-

◆ 7. Advantages

- High security compared to ECB
- Prevents identical ciphertext patterns

- **Widely used in secure communication**
-

◆ 8. Disadvantages

- **Cannot be parallelized (slow)**
- **Error in one block affects next block**
- **Requires secure IV generation**

Q2. What is substitution cipher? Explain the types of mono alphabetic cipher poly alphabetic cipher with suitable example

ANS:

Substitution Cipher

◆ 1. Introduction

- **A substitution cipher is a cryptographic technique in which each character of plaintext is replaced (substituted) with another character.**
 - **The positions of characters are not changed, only the symbols are replaced.**
 - **It is one of the basic classical encryption techniques.**
-

◆ 2. Definition

- **A substitution cipher is a method of encryption where each letter (or group of letters) in the plaintext is systematically replaced by another letter to produce ciphertext.**
-

◆ 3. Types of Substitution Cipher

◆ 1. Monoalphabetic Cipher

◆ Explanation

- In monoalphabetic cipher, each letter of plaintext is replaced by a fixed corresponding letter.
- The substitution mapping is one-to-one and constant throughout the message.

👉 Same letter always maps to same cipher letter.

◆ Example

Plaintext Alphabet: A B C D E F

Cipher Alphabet: D E F G H I

If plaintext = HELLO

- H → K
- E → H
- L → O
- L → O
- O → R

👉 Ciphertext = KHOOR

◆ Characteristics

- Simple encryption
 - Easy to break using frequency analysis
 - Not highly secure
-

◆ 2. Polyalphabetic Cipher

◆ Explanation

- In polyalphabetic cipher, a letter can be encrypted into different ciphertext letters depending on position.
 - Uses multiple substitution alphabets.
 - Increases security compared to monoalphabetic cipher.
-

◆ Example (Vigenère Cipher)

Plaintext: HELLO

Key: KEYKE

Now encrypt using shifting:

- $H + K = R$
- $E + E = I$
- $L + Y = J$
- $L + K = V$
- $O + E = S$

👉 Ciphertext = RIJVS

◆ Characteristics

- More secure than monoalphabetic cipher
- Same letter can have different ciphertext values
- Harder to break using frequency analysis

◆ 4. Difference Between Monoalphabetic and Polyalphabetic Cipher

Feature	Monoalphabetic	Polyalphabetic
Substitution	Fixed	Multiple
Security	Low	High
Repetition	Same mapping	Changes every time
Example	Caesar cipher	Vigenère cipher

Q3. Explain different types of attacks on message integrity and availability

ANS:

Attacks on Message Integrity and Availability

◆ 1. Introduction

- In network security, attackers target two important properties:
 - Integrity → Ensures data is not modified or tampered.
 - Availability → Ensures services/data are accessible when needed.
- Attacks are designed either to alter data (integrity attacks) or to disrupt services (availability attacks).

◆ 2. Attacks on Message Integrity

◆ 1. Modification Attack

- In this attack, the attacker changes the actual content of the message during transmission.
- The receiver gets altered or incorrect data without knowing it.

👉 Example: Changing “₹5000 transfer” to “₹50000 transfer”.

◆ 2. Fabrication Attack

- Attacker creates fake messages and sends them to the receiver.
- The fake message appears as if it came from a legitimate sender.

👉 Example: Fake login request or fake transaction request.

◆ 3. Replay Attack

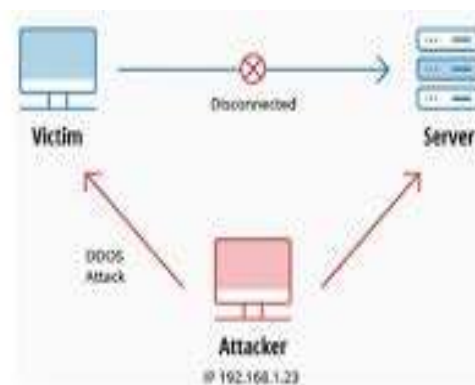
- Attacker captures a valid message and retransmits it later.
- System treats it as a new legitimate request.

👉 Example: Reusing old authentication token to log in again.

◆ 4. Man-in-the-Middle (MITM) Attack

- Attacker secretly intercepts communication between two parties.
 - Can read, modify, or inject messages without detection.
-

🖥️ Diagram: MITM Attack



◆ 5. Masquerading (Impersonation Attack)

- Attacker pretends to be a legitimate user.
- Gains unauthorized access to system.

◆ 3. Attacks on Message Availability

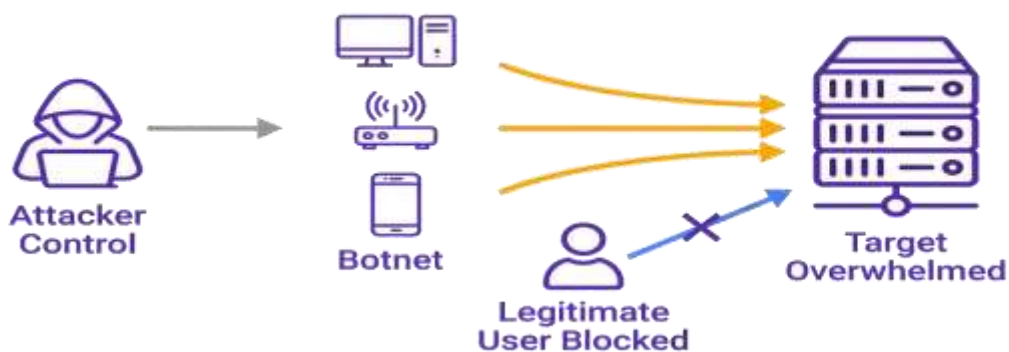
◆ 1. Denial of Service (DoS) Attack

- Attacker floods a system with excessive requests or traffic.
 - System becomes slow or completely unavailable.
-

◆ 2. Distributed Denial of Service (DDoS) Attack

- Multiple compromised systems (botnet) attack a target simultaneously.
 - Causes server overload and shutdown.
-

 Diagram: DDoS Attack



◆ 3. Network Congestion Attack (Flooding)

- Attacker sends large volume of packets continuously.
 - Consumes bandwidth and blocks legitimate traffic.
-

◆ 4. Resource Exhaustion Attack

- **Attacker consumes system resources like:**
 - **CPU**
 - **Memory**
 - **Disk space**
 - **System becomes unable to serve users.**
-

♦ **5. Jamming Attack (Wireless Networks)**

- **Attacker interferes with wireless signals using noise.**
 - **Communication between nodes gets disrupted.**
-

♦ **6. Blackhole Attack (Adhoc Networks)**

- **Malicious node drops all packets instead of forwarding them.**
 - **Leads to data loss and communication failure.**
-

♦ **4. Effects of These Attacks**

Integrity Attacks:

- **Data corruption**
- **Fake or modified information**
- **Loss of trust in communication**

Availability Attacks:

- **Service downtime**
 - **Network congestion**
 - **Financial loss**
 - **System crash**
-

◆ 5. Prevention Methods

◆ For Integrity:

- Encryption (AES, RSA)
- Hashing (SHA)
- Digital signatures
- Authentication mechanisms

◆ For Availability:

- Firewalls and IDS/IPS
- Rate limiting
- Load balancing
- Anti-DDoS protection systems

Q4. What is network attack? Differentiate between active and passive attack

ANS:

Network Attack and Types: Active vs Passive Attack

◆ 1. Network Attack (Meaning)

◆ Definition

A network attack is any deliberate attempt by an unauthorized user (attacker) to access, disrupt, modify, or destroy data or network services.

◆ Explanation (Deep)

- Networks are designed to ensure confidentiality, integrity, and availability (CIA).
- Attackers try to break these security properties.

- **Network attacks can target:**
 - **Data (messages, files)**
 - **Devices (routers, servers)**
 - **Services (websites, applications)**
-

◆ **Goals of Attacker**

- **Steal sensitive information**
 - **Disrupt communication**
 - **Modify data**
 - **Make services unavailable**
-

◆ **Classification**

Network attacks are mainly divided into:

- **Passive Attack**
- **Active Attack**

No.	Passive Attack	Active Attack
1	It is an attack where attacker only listens to communication	It is an attack where attacker modifies or disrupts data
2	No change in original data	Data is altered, deleted, or injected
3	Hard to detect	Easier to detect
4	Affects confidentiality	Affects integrity and availability
5	Attacker is silent observer	Attacker actively participates
6	Does not harm system resources directly	Can overload or damage system

No.	Passive Attack	Active Attack
7	Example: Eavesdropping	Example: DoS, MITM attack
8	Used to steal information secretly	Used to damage or control system
9	Low risk but dangerous due to secrecy	High risk and more destructive
10	Prevention: Encryption	Prevention: Authentication, firewall, IDS

Q5. What is the importance & need for security and explain network attack

ANS:

Importance and Need for Security & Network Attacks

◆ **1. Importance and Need for Security**

◆ **1. Introduction**

- Network security is required to protect data and communication systems from unauthorized access, misuse, and attacks.
- As networks grow (internet, mobile, cloud), security becomes very important.

◆ **2. Need for Security (Deep Points)**

1. Protection of Confidential Data

- Prevents unauthorized users from accessing sensitive information like passwords, banking data, etc.

2. Data Integrity

- Ensures data is not modified or tampered during transmission.
-

3. Availability of Services

- Ensures network services are always accessible to users.
-

4. Authentication

- Confirms the identity of users before granting access.
-

5. Prevention of Cyber Attacks

- Protects against attacks like hacking, malware, DoS, phishing.
-

6. Secure Communication

- Ensures safe communication over insecure networks like the internet.
-

7. Protection of Resources

- Prevents misuse of network resources like CPU, bandwidth, storage.
-

8. Privacy Protection

- Protects personal user information from leakage.
-

9. Trust in System

- Builds trust between users and service providers.
-

10. Legal & Financial Safety

- Prevents financial loss and legal issues due to data breaches.

➤ Network Attack (Meaning)

◆ Definition

A network attack is any deliberate attempt by an unauthorized user (attacker) to access, disrupt, modify, or destroy data or network services.

◆ Explanation (Deep)

- Networks are designed to ensure confidentiality, integrity, and availability (CIA).
 - Attackers try to break these security properties.
 - Network attacks can target:
 - Data (messages, files)
 - Devices (routers, servers)
 - Services (websites, applications)
-

◆ Goals of Attacker

- Steal sensitive information
- Disrupt communication
- Modify data
- Make services unavailable

Types of Network Attacks

1. Passive Attack

- Attacker only listens or monitors communication
 - No modification of data
 - Example: Eavesdropping, traffic analysis
-

2. Active Attack

- Attacker modifies, deletes, or injects data
- Affects system operation
- Example: DoS, MITM, replay attack

Q6. Explain different block cipher muscles

ANS:

Block Cipher Modes of Operation (Deep Explanation)

◆ 1. Introduction

- A block cipher encrypts data in fixed-size blocks (example: 64-bit or 128-bit).
 - Real messages are longer than one block, so we divide them into multiple blocks.
 - To securely encrypt these multiple blocks, we use Block Cipher Modes of Operation.
 - These modes define how each block is encrypted using the cipher and how blocks are linked.
-

◆ 2. Need of Block Cipher Modes

- Prevent repetition of patterns in ciphertext

- Improve security of encryption
- Handle messages of arbitrary length
- Provide confidentiality in communication

◆ 3. Types of Block Cipher Modes

◆ 1. Electronic Code Book (ECB) Mode

◆ Concept (Deep)

- ECB is the simplest mode.
 - Each plaintext block is encrypted independently using the same key.
 - There is no interaction between blocks.
-

◆ Working

- $P1 \rightarrow \text{encrypt} \rightarrow C1$
 - $P2 \rightarrow \text{encrypt} \rightarrow C2$
 - $P3 \rightarrow \text{encrypt} \rightarrow C3$
-

◆ Diagram

$P1 \rightarrow E \rightarrow C1$

$P2 \rightarrow E \rightarrow C2$

$P3 \rightarrow E \rightarrow C3$

◆ Characteristics

- Very fast

- Supports parallel encryption
 - Same plaintext → same ciphertext (big weakness)
 - Not secure for real applications
-

◆ Weakness

- Reveals patterns in data (e.g., images, text repetition)
-

◆ 2. Cipher Block Chaining (CBC) Mode

◆ Concept (Deep)

- CBC improves ECB by adding dependency between blocks.
 - Each plaintext block is XORed with previous ciphertext block before encryption.
 - First block uses Initialization Vector (IV).
-

◆ Working Formula

- $C1 = E(P1 \oplus IV)$
 - $C2 = E(P2 \oplus C1)$
 - $C3 = E(P3 \oplus C2)$
-

◆ Characteristics

- Strong security
 - Hides patterns in plaintext
 - Sequential processing (no parallel encryption)
 - Error in one block affects next block
-

◆ Advantage

- Most widely used mode in secure systems
-

◆ 3. Cipher Feedback (CFB) Mode

◆ Concept (Deep)

- Converts block cipher into a self-synchronizing stream cipher.
 - Encryption of previous ciphertext generates keystream.
-

◆ Working

- $C1 = P1 \oplus E(IV)$
- $C2 = P2 \oplus E(C1)$
- $C3 = P3 \oplus E(C2)$

◆ Characteristics

- Self-synchronizing
 - Good for real-time transmission
 - Error affects current and next blocks
 - No need for padding
-

◆ 4. Output Feedback (OFB) Mode

◆ Concept (Deep)

- Similar to CFB but keystream is independent of plaintext and ciphertext.
 - Only encryption function is used to generate keystream.
-

◆ Working

- $K1 = E(IV)$
 - $C1 = P1 \oplus K1$
 - $K2 = E(K1)$
 - $C2 = P2 \oplus K2$
-

◆ Characteristics

- No error propagation
 - Keystream independent of data
 - Suitable for noisy channels
 - Not self-synchronizing
-

◆ 4. Comparison of Modes

Feature	ECB	CBC	CFB	OFB
Security	Low	High	Medium	Medium
Pattern hiding	No	Yes	Yes	Yes
Parallel processing	Yes	No	No	No
Error propagation	No	Yes	Yes	No
Usage	Rare	Very common	Stream encryption	Noisy channels

Q7. Explain distributed denial of service attacks in details

ANS:

Distributed Denial of Service (DDoS) Attack

◆ 1. Introduction

- **A Distributed Denial of Service (DDoS) attack is a type of cyber attack where multiple compromised systems are used to flood a target system with excessive traffic.**
 - **The main goal is to make a service unavailable to legitimate users.**
 - **It is one of the most dangerous attacks in network security because it uses a large number of attacking machines (botnet).**
-

◆ 2. Definition

- **A DDoS attack is an attack in which many infected computers (bots/zombies) send massive traffic to a target system simultaneously, causing overload and service disruption.**
-

◆ 3. Working of DDoS Attack (Deep Explanation)

◆ Step 1: Botnet Creation

- **Attacker infects multiple computers using malware/viruses/trojans.**
 - **These infected devices are called bots or zombies.**
 - **A group of bots is called a botnet.**
-

◆ Step 2: Command and Control (C&C)

- **Attacker controls bots using a central command server.**
 - **Sends instructions to all bots to start attack.**
-

◆ Step 3: Attack Execution

- All bots send huge number of requests/packets to the target server simultaneously.
 - Traffic may include fake requests, packets, or service requests.
-

◆ Step 4: System Overload

- Target server resources like:
 - CPU
 - Memory
 - Bandwidth get exhausted.
-

◆ Step 5: Service Disruption

- Server becomes:
 - Slow
 - Unresponsive
 - Completely down

◆ 4. Diagram of DDoS Attack



◆ 5. Types of DDoS Attacks

◆ 1. Volume-Based Attacks

- Focus on flooding bandwidth with huge traffic.
 - Example: UDP flood, ICMP flood
-

◆ 2. Protocol Attacks

- Exploit weaknesses in network protocols.
 - Example: SYN flood attack
 - Goal: Exhaust server resources.
-

◆ 3. Application Layer Attacks

- Target web applications (HTTP, web servers).
 - Example: HTTP flood
 - Hard to detect because traffic looks normal.
-

◆ 6. Effects of DDoS Attack

- Website or service becomes unavailable
 - Network congestion
 - Financial loss for companies
 - Reputation damage
 - System crash or slowdown
-

◆ 7. Prevention Methods

◆ 1. Firewalls

- Filter malicious traffic.
 - ◆ 2. Intrusion Detection System (IDS)
 - Detect abnormal traffic patterns.
 - ◆ 3. Rate Limiting
 - Limits number of requests from a single IP.
 - ◆ 4. Load Balancing
 - Distributes traffic across multiple servers.
 - ◆ 5. Anti-DDoS Services
 - Cloud-based protection systems (CDN security).
-

- ◆ 8. Advantages (Attacker's View)
 - Easy using botnets
 - Hard to trace attacker
 - Very effective in crashing systems
-

- ◆ 9. Disadvantages (Victim's Impact)
 - Service downtime
 - Loss of revenue
 - Data unavailability
 - Reputation damage

Q8. What is stream cipher? Explain encryption process using stream cipher with suitable example

ANS:

Stream Cipher (Deep Explanation)

◆ 1. Introduction

- A stream cipher is a symmetric key encryption technique in which plaintext is encrypted one bit or one byte at a time.
 - Instead of dividing data into blocks, it processes data as a continuous stream.
 - It uses a keystream generator that produces a pseudo-random sequence of bits.
 - Encryption is done using XOR operation between plaintext and keystream.
-

◆ 2. Definition

- A stream cipher is a cryptographic algorithm in which plaintext is combined with a pseudo-random keystream bit-by-bit using XOR operation to produce ciphertext.
-

◆ 3. Core Principle

✓ Encryption Formula:

- $C = P \oplus K$

✓ Decryption Formula:

- $P = C \oplus K$

Where:

- P = Plaintext

- **C = Ciphertext**
 - **K = Keystream**
-

◆ **4. Architecture / Working of Stream Cipher**

◆ **Components:**

1. **Secret Key**
 2. **Key Stream Generator (PRNG)**
 3. **XOR Operation Unit**
-

◆ **Step-by-Step Process**

◆ **Step 1: Key Input**

- **A secret key is given to both sender and receiver.**
-

◆ **Step 2: Keystream Generation**

- **Key is fed into a pseudo-random number generator (PRNG).**
 - **It produces a long sequence of random bits called keystream.**
-

◆ **Step 3: Encryption**

- **Each bit/byte of plaintext is XORed with keystream bit.**
 - **Output is ciphertext.**
-

◆ **Step 4: Transmission**

- **Ciphertext is sent over insecure channel.**
-

◆ Step 5: Decryption

- Receiver generates same keystream using same key.
 - XOR with ciphertext gives original plaintext.
-

◆ 5. Diagram of Stream Cipher

◆ Encryption

Plaintext (P) ———┐
 XOR —→ Ciphertext (C)
Keystream (K) ———┘

◆ Decryption

Ciphertext (C) ———┐
 XOR —→ Plaintext (P)
Keystream (K) ———┘

◆ 6. Example (Binary)

Let:

- Plaintext (P) = 11001010
 - Keystream (K) = 10110101
-

◆ Encryption:

P = 11001010

K = 10110101

-----XOR

C = 01111111

◆ **Decryption:**

C = 01111111

K = 10110101

----- XOR

P = 11001010

✓ Original data is recovered.

◆ **7. Types of Stream Cipher**

◆ **1. Synchronous Stream Cipher**

- **Keystream is generated independently of plaintext and ciphertext.**
 - **Both sender and receiver must stay synchronized.**
 - **Example: RC4 (conceptually)**
-

◆ **2. Self-Synchronizing Stream Cipher**

- **Keystream depends on previous ciphertext bits.**
 - **Automatically resynchronizes after errors.**
-

◆ **8. Characteristics of Stream Cipher**

- **Encrypts data bit-by-bit or byte-by-byte**
 - **Uses XOR operation**
 - **Very fast and efficient**
 - **No padding required**
 - **Suitable for real-time systems**
-

◆ 9. Advantages

- High speed encryption
 - Low hardware complexity
 - Suitable for real-time communication (voice/video)
 - Minimal delay
-

◆ 10. Disadvantages

- Weak if keystream is reused (very dangerous)
 - Error in transmission affects output
 - Requires secure key generation
 - Less secure than modern block cipher systems in some cases
-

◆ 11. Applications

- Wireless communication (Wi-Fi, Bluetooth)
- Secure voice calls
- Streaming data encryption
- Mobile communication systems

Q9. What is Cipher Block Chaining (CBC)? Explain the process of CBC with suitable diagram

ANS:

Cipher Block Chaining (CBC) Mode

◆ 1. Introduction

- CBC (Cipher Block Chaining) is an important block cipher mode of operation used in cryptography to improve security.
 - It overcomes the weakness of ECB by ensuring that identical plaintext blocks produce different ciphertext blocks.
 - This is achieved by making each block dependent on the previous ciphertext block.
-

◆ 2. Definition

- CBC is a block cipher mode in which each plaintext block is first XORed with the previous ciphertext block (or IV for first block) and then encrypted using a secret key.
-

◆ 3. Working Principle of CBC

◆ Basic Idea

- Each encryption depends on:
 - Current plaintext block
 - Previous ciphertext block
 - This creates a chain of encryption, hence called “chaining mode”.
-

◆ 4. Encryption Process (Step-by-Step)

Let:

- P_1, P_2, P_3 = plaintext blocks
 - C_1, C_2, C_3 = ciphertext blocks
 - IV = Initialization Vector
 - E = Encryption function
-

♦ Step 1: First Block Encryption

- First plaintext block is XORed with IV:

👉 $C_1 = E(P_1 \oplus IV)$

♦ Step 2: Second Block Encryption

- Second block is XORed with previous ciphertext:

👉 $C_2 = E(P_2 \oplus C_1)$

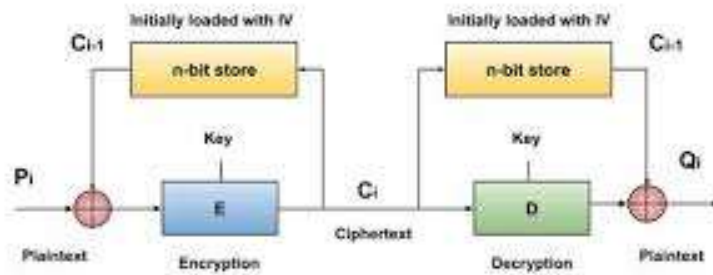
♦ Step 3: Third Block Encryption

👉 $C_3 = E(P_3 \oplus C_2)$

♦ General Formula

👉 $C_i = E(P_i \oplus C_{i-1})$

♦ 5. Diagram of CBC Mode



◆ 6. Decryption Process

◆ Idea

- Decryption is reverse of encryption.

Formula:

👉 $P_i = D(C_i) \oplus C_{i-1}$

Where:

- D = Decryption function
- C_{i-1} = previous ciphertext block
- For first block, IV is used instead of C_0

◆ Steps:

1. Decrypt ciphertext block using key
2. XOR result with previous ciphertext block
3. Get original plaintext

◆ 7. Why IV is Important?

- IV ensures randomness in encryption
- Even if same plaintext is encrypted multiple times, ciphertext will be different

- Prevents pattern analysis attack
-

◆ 8. Characteristics of CBC Mode

◆ 1. Chaining Property

- Each block depends on previous ciphertext → strong dependency
-

◆ 2. High Security

- Eliminates repetition patterns (unlike ECB)
-

◆ 3. Sequential Processing

- Cannot encrypt blocks in parallel
 - Slower than ECB
-

◆ 4. Error Propagation

- If one ciphertext block is corrupted:
 - Current plaintext block is corrupted
 - Next block is also affected
-

◆ 5. Requires IV

- IV must be random and unique for security
-

◆ 9. Advantages

- Very strong security
- Prevents pattern leakage

- Widely used in SSL/TLS, file encryption
-

◆ 10. Disadvantages

- No parallel encryption (slow)
 - Error propagation
 - Needs secure IV management
-

◆ 11. Real-World Applications

- Secure web communication (HTTPS)
- VPN encryption
- File encryption systems
- Banking transactions

**Q10. Describe the following network security threats. i) Unauthorized access
ii) Man in the middle**

ANS: Network Security Threats

i) Unauthorized Access

◆ 1. Definition

- Unauthorized access is a security threat where an attacker gains access to a network, system, or data without permission or authentication.
-

◆ 2. Explanation

- In secure systems, users must pass authentication (username/password, OTP, etc.).
- If an attacker bypasses this process, it is called unauthorized access.

- It leads to data theft, misuse, or modification of information.
-

◆ 3. How it Happens

- Weak passwords
 - Stolen credentials
 - Brute force attacks
 - Exploiting system vulnerabilities
 - Phishing attacks
-

◆ 4. Effects

- Data leakage
- Financial loss
- Privacy violation
- System misuse

◆ 5. Example

- Hacker logging into a bank account using stolen credentials.
-

◆ 6. Prevention

- Strong passwords
 - Multi-factor authentication (MFA)
 - Firewalls
 - Intrusion Detection Systems (IDS)
 - Regular security updates
-

ii) Man-in-the-Middle (MITM) Attack

◆ 1. Definition

- A Man-in-the-Middle attack is a network attack where an attacker secretly intercepts and possibly modifies communication between two parties without their knowledge.
-

◆ 2. Explanation

- Two users (A and B) think they are communicating directly.
 - Attacker secretly sits in between and:
 - Reads messages
 - Modifies data
 - Sends fake responses
 - Both users remain unaware of attack.
-

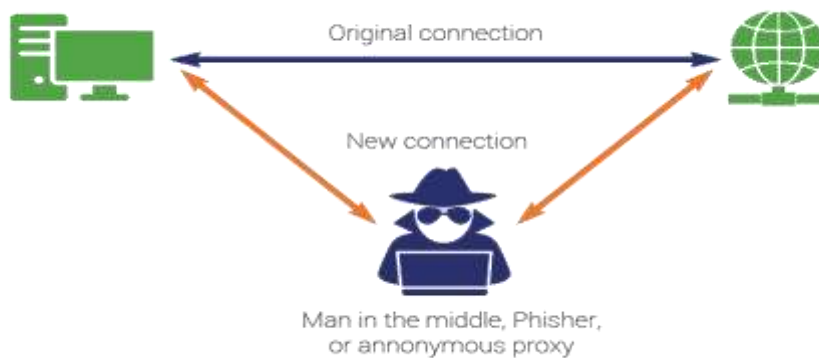
◆ 3. Working of MITM Attack

1. Attacker intercepts communication channel
 2. Acts as a bridge between sender and receiver
 3. Captures sensitive data (passwords, OTPs, etc.)
 4. May alter or inject false messages
-

◆ 4. Diagram of MITM

A <-----> Attacker <-----> B

(intercepts & modifies data)



◆ 5. Effects

- Data theft (passwords, banking info)
- Identity theft
- Data modification
- Loss of trust in communication

◆ 6. Example

- Attacker intercepting Wi-Fi communication in public network and stealing login credentials.

◆ 7. Prevention

- Use HTTPS (secure websites)
- Encryption (SSL/TLS)
- VPN for secure communication
- Digital certificates
- Strong authentication mechanisms

Q11. Describe the following fundamental principles of Information security [6]

i) Integrity ii) Authentication iii) Authorization and Access Control

ANS:

Fundamental Principles of Information Security (Deep Explanation)

Information security is built on core principles that ensure safe communication and protect data from unauthorized actions. The three important principles are:

◆ i) Integrity

◆ Definition

- Integrity ensures that information remains accurate, complete, and unaltered during storage, processing, and transmission unless changed by an authorized user.

◆ Deep Explanation

- Integrity focuses on protecting data from unauthorized modification, deletion, or creation.
- It ensures that the received data is exactly the same as sent data.
- Even a small change in data should be detectable.
- Integrity is important in systems where data accuracy is critical (banking, medical records, transactions).

◆ Threats to Integrity

- Data tampering by attackers
- Man-in-the-Middle (MITM) attacks
- Virus or malware modification
- Replay attacks (sending old messages again)

◆ Mechanisms to Ensure Integrity

- Hash functions (SHA, MD5) → generate fixed digest
- Digital signatures → verify data source and correctness
- Checksums → detect transmission errors
- MAC (Message Authentication Code) → ensures authenticity + integrity

◆ Example

- In online banking, if a user sends “₹5000 transfer”, it must not be changed to “₹50000” during transmission.

◆ Importance

- Prevents data corruption
- Ensures trust in communication systems
- Critical for financial and medical systems

◆ ii) Authentication

◆ Definition

- Authentication is the process of verifying the identity of a user, device, or system before granting access to resources.

◆ Deep Explanation

- It answers the question: “Who are you?”
- Only authenticated users are allowed to access system resources.
- Authentication happens before authorization.

- It ensures that attackers cannot impersonate legitimate users.
-

◆ Types of Authentication

1. Knowledge-based

- Passwords, PINs

2. Possession-based

- OTP, smart cards, security tokens

3. Biometric-based

- Fingerprint, face recognition, iris scan

4. Multi-Factor Authentication (MFA)

- Combination of two or more methods (e.g., password + OTP)
-

◆ Authentication Process

1. User enters credentials
 2. System verifies with database
 3. If valid → access granted
 4. If invalid → access denied
-

◆ Example

- Logging into Gmail using password + OTP verification.
-

◆ Importance

- Prevents unauthorized access
- Protects sensitive data
- Ensures system security before granting access

◆ iii) Authorization and Access Control

◆ Definition

- Authorization determines what actions a user is allowed to perform after authentication.
- Access Control is the mechanism that enforces these permissions in a system.

◆ Deep Explanation

- After identity verification (authentication), system checks user privileges.
- It defines access rules such as:
 - Read access
 - Write access
 - Delete access
 - Admin privileges
- It ensures users only access authorized resources.

◆ Access Control Process

1. User is authenticated
2. System checks user role/permissions
3. Access is granted or denied based on policy

◆ Types of Access Control

1. Role-Based Access Control (RBAC)

- Access based on user role (Admin, Student, User)

2. Mandatory Access Control (MAC)

- System-enforced rules (high security environments)

3. Discretionary Access Control (DAC)

- Owner decides who can access data
-

♦ Example

- Admin can delete or modify database
 - Normal user can only view data
-

♦ Importance

- Prevents misuse of system resources
- Ensures least privilege principle
- Improves overall system security

Q12. What is Cipher Feedback Mode(CFM) and Electronic Code book (ECB)?

ANS:

Electronic Code Book (ECB) Mode – Deep Theory

♦ 1. Conceptual Idea

- ECB is the most basic block cipher mode of operation.
- It treats the plaintext message as a sequence of independent fixed-size blocks.
- Each block is encrypted separately using the same encryption key.

👉 So, ECB behaves like a “code book” where each plaintext block has a fixed ciphertext mapping.

◆ 2. Working Philosophy

- The main idea is independence of blocks.
- There is no chaining or dependency between blocks.
- Encryption is purely:
 - Input block → Encryption function → Output block

This makes ECB very simple but also cryptographically weak.

◆ 3. Mathematical Representation

- Encryption: $C_i = E(K, P_i)$
- Decryption: $P_i = D(K, C_i)$

Where:

- P_i = plaintext block
 - C_i = ciphertext block
 - K = secret key
-

◆ 4. Internal Behavior

- Each block goes through the same algorithm (like AES or DES).
 - Same input block always produces same output block.
 - No randomness is introduced.
-

◆ 5. Security Weakness (Very Important Theory Point)

- Since mapping is fixed:
 - Repeated plaintext blocks → repeated ciphertext blocks
- This leads to pattern leakage attack.

👉 **Example:**

If plaintext is image data, ECB will still show visible structure after encryption.

◆ **6. Cryptographic Limitation**

- **ECB does not provide semantic security.**
 - **It fails against:**
 - **statistical analysis**
 - **pattern recognition attacks**
-

◆ **7. Conclusion (ECB)**

- **ECB is simple and fast but cryptographically insecure.**
 - **It is mainly used only for small or random data, not secure communication.**
-
-

Cipher Feedback Mode (CFB) – Deep Theory

◆ **1. Conceptual Idea**

- **CFB is a self-synchronizing stream cipher mode derived from block cipher.**
- **It converts a block cipher into a stream cipher by introducing feedback of ciphertext.**

👉 **Main idea:**

Instead of encrypting plaintext directly, we encrypt previous ciphertext (or IV) and use it as a keystream.

◆ 2. Working Philosophy

- Encryption depends on:
 - Current plaintext
 - Output of previous encryption

This creates a chain of dependency, increasing security.

◆ 3. Mathematical Representation

- $C1 = P1 \oplus E(K, IV)$
 - $C2 = P2 \oplus E(K, C1)$
 - $Ci = Pi \oplus E(K, Ci-1)$
-

◆ 4. Internal Mechanism (Deep Flow)

1. IV is encrypted using block cipher
2. Output acts as keystream
3. Keystream is XORed with plaintext
4. Ciphertext is generated
5. Ciphertext is fed back into next step

👉 This is called feedback mechanism.

◆ 5. Why CFB is Secure

- Even if same plaintext is repeated:
 - ciphertext changes due to feedback
- No fixed mapping like ECB
- Introduces randomness via IV + chaining

◆ 6. Self-Synchronization Property

- If some bits are lost during transmission:
 - system automatically resynchronizes after few blocks
 - This makes CFB useful in noisy communication channels
-

◆ 7. Error Propagation Behavior

- If one ciphertext bit is corrupted:
 - It affects current plaintext block
 - Also affects next block partially
 - But error does not propagate indefinitely.
-

◆ 8. Cryptographic Strength

- Provides better confidentiality than ECB
 - Resistant to pattern analysis
 - Suitable for streaming and real-time systems
-

◆ 9. Conclusion (CFB)

- CFB is a strong, secure, feedback-based mode that improves security by removing pattern repetition and introducing dependency between blocks.

Q13. Explain the process of CFM with suitable diagram.

ANS:

Cipher Feedback Mode (CFB / CFM) – Theory

◆ 1. Introduction

- **Cipher Feedback Mode (CFB) is a block cipher mode of operation in cryptography.**
 - **It is designed to convert a block cipher into a stream cipher, allowing encryption of data in smaller units like bits or bytes instead of fixed-size blocks.**
 - **It introduces a feedback mechanism, where previously generated ciphertext is used in the encryption of the next plaintext segment.**
-

◆ 2. Concept

- **In CFB mode, encryption does not directly operate on plaintext blocks.**
 - **Instead, the output of the encryption function is combined with plaintext using XOR operation.**
 - **The input to the encryption function is either:**
 - **Initialization Vector (IV) for the first block**
 - **Previous ciphertext block for subsequent blocks**
-

◆ 3. Working Principle (Theory)

- **The sender and receiver share a secret key and an Initialization Vector (IV).**
- **The IV is encrypted using the block cipher algorithm.**
- **The resulting output acts as a keystream generator.**
- **This keystream is XORed with plaintext to generate ciphertext.**
- **The ciphertext is then fed back into the system for further encryption.**

◆ 4. Mathematical Representation

- $C_1 = P_1 \oplus E(IV, K)$
- $C_2 = P_2 \oplus E(C_1, K)$
- $C_i = P_i \oplus E(C_{i-1}, K)$

Where:

- P_i = plaintext block
 - C_i = ciphertext block
 - K = secret key
 - IV = initialization vector
 - E = encryption function
-

◆ 5. Nature of CFB Mode

- It behaves like a self-synchronizing stream cipher.
 - Encryption and decryption use the same encryption algorithm, making implementation efficient.
 - It ensures that each ciphertext block depends on previous ciphertext, creating a chain of dependency.
-

◆ 6. Security Features

- Provides better security than ECB mode because it removes repetition patterns.
 - The use of IV ensures that the same plaintext produces different ciphertext each time.
 - The feedback mechanism makes statistical attacks difficult.
-

◆ 7. Error Behavior

- A single-bit error in ciphertext affects:
 - The corresponding plaintext block
 - Part of the next block
 - After a few blocks, synchronization is automatically restored.
-

◆ 8. Key Characteristics

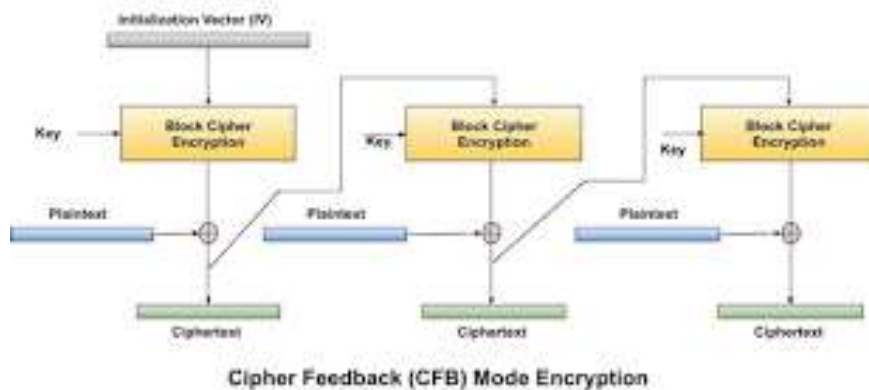
- Converts block cipher into stream cipher
- Uses feedback from previous ciphertext
- Does not require padding
- Sequential processing (no parallel encryption)
- Self-synchronizing in nature

◆ 9. Advantages

- Stronger security than ECB
 - Suitable for real-time communication systems
 - No need for padding
 - Works well in noisy transmission environments
-

◆ 10. Disadvantages

- Cannot be parallelized
- Slower than ECB mode
- Error propagation occurs
- Requires proper IV management



Q14. Explain the encryption process using stream cipher with suitable example

ANS:

Stream Cipher – Encryption Process (Deep Theory)

◆ 1. Introduction

- A stream cipher is a symmetric key encryption technique where plaintext is encrypted continuously one bit or one byte at a time.
 - Unlike block ciphers, it does not divide data into fixed-size blocks.
 - It generates a keystream (pseudo-random bit sequence) which is combined with plaintext using XOR operation.
-

◆ 2. Core Idea

- The main idea of stream cipher is:

“Security is achieved by mixing plaintext with a random-looking keystream.”

- If keystream is truly random and never reused, encryption becomes very strong.
-

◆ 3. Components of Stream Cipher System

1. Plaintext (P) – original message
 2. Secret Key (K) – shared between sender and receiver
 3. Keystream Generator (PRNG) – produces pseudo-random bits
 4. XOR Operation Unit – performs encryption
-

◆ 4. Encryption Process (Deep Step-by-Step)

◆ Step 1: Key Input

- A secret key is provided to the keystream generator.
 - The key acts as a seed for pseudo-random generation.
-

◆ Step 2: Keystream Generation

- A Pseudo Random Number Generator (PRNG) produces a keystream:
 - K1, K2, K3, ...
- Keystream length matches plaintext length.

👉 Important:

Keystream is not truly random, but appears random mathematically.

◆ Step 3: XOR Encryption

- Each bit/byte of plaintext is XORed with keystream:

👉 $C_i = P_i \oplus K_i$

- XOR rule:
 - $0 \oplus 0 = 0$
 - $1 \oplus 1 = 0$
 - $0 \oplus 1 = 1$
 - $1 \oplus 0 = 1$

◆ Step 4: Ciphertext Generation

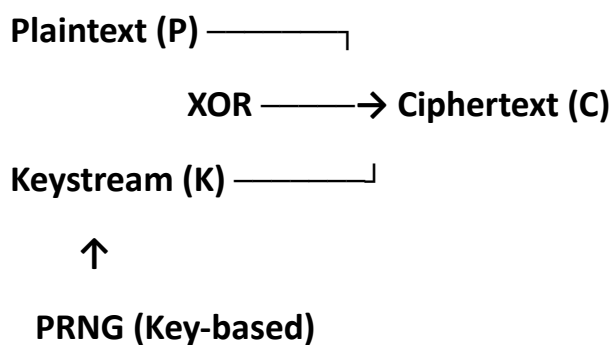
- Output of XOR is ciphertext stream.
- Ciphertext is transmitted over insecure channel.

◆ Step 5: Decryption Process

- Same keystream is generated using same key.
- XOR again retrieves original plaintext:

👉 $P_i = C_i \oplus K_i$

◆ 5. Diagram (Stream Cipher Encryption)



◆ 6. Detailed Working Logic

- Encryption depends entirely on:
 - Key
 - Keystream generator
 - There is no block structure, so:
 - No padding required
 - Works in real-time
-

◆ 7. Example (Binary Deep Explanation)

Let:

- Plaintext (P) = 11010101
- Keystream (K) = 10111010

◆ Encryption:

P = 11010101

K = 10111010

----- XOR

C = 01101111

◆ Decryption:

C = 01101111

K = 10111010

----- XORP = 11010101

✓ Original message is recovered perfectly.

◆ 8. Why XOR Works (Important Theory Point)

- XOR is reversible operation:
 - $A \oplus B \oplus B = A$
- So same keystream used twice cancels out encryption.

◆ 9. Types of Stream Cipher

◆ 1. Synchronous Stream Cipher

- Keystream depends only on key
 - Sender and receiver must stay synchronized
-

◆ 2. Self-Synchronizing Stream Cipher

- Keystream depends on previous ciphertext
 - Can recover automatically after error
-

◆ 10. Security Aspects

- Security depends on:
 - Strength of PRNG
 - Unpredictability of keystream
 - If keystream is reused → serious vulnerability (key reuse attack)
-

◆ 11. Advantages

- Very fast encryption/decryption
 - Suitable for real-time systems
 - No padding required
 - Simple hardware implementation
-

◆ 12. Disadvantages

- Weak if keystream is reused
 - Error affects output bits
 - Requires secure key management
 - Less robust than modern block ciphers in some cases
-

◆ 13. Applications

- Secure voice communication
- Wireless networks (Wi-Fi, Bluetooth)
- Video streaming encryption
- Mobile communication systems

Q15. Comment on security principles and security services

ANS:

Security Principles and Security Services

◆ 1. Security Principles (Fundamental Principles)

◆ Introduction

Security principles are the basic rules used to protect information and systems from unauthorized access, modification, and destruction. They form the foundation of network and information security.

◆ 1. Confidentiality

- Ensures that information is accessible only to authorized users.
- Prevents unauthorized disclosure of data.
- Achieved using encryption techniques (AES, RSA).

👉 Example: Passwords and banking data are kept secret.

◆ 2. Integrity

- Ensures that data is accurate, complete, and not modified illegally.
- Any unauthorized change must be detectable.

- Achieved using hash functions and digital signatures.

👉 Example: Transaction amount should not be changed during transfer.

◆ 3. Availability

- Ensures that systems and data are available whenever needed by authorized users.
- Protects against attacks like DoS and system failure.

👉 Example: Website should remain accessible 24/7.

◆ 4. Authentication

- Verifies the identity of a user or system.
- Ensures that the user is genuine.

👉 Example: Login using username, password, OTP.

◆ 5. Authorization

- Defines what actions a user is allowed to perform after authentication.
- Controls access levels (read, write, delete).

👉 Example: Admin can delete data, user cannot.

◆ 6. Non-repudiation

- Ensures that sender cannot deny sending a message.
- Provides proof of origin.
- Achieved using digital signatures.

◆ 2. Security Services

◆ Introduction

Security services are the mechanisms provided by a system to implement security principles. they are defined in OSI security architecture.

◆ 1. Authentication Service

- Confirms identity of users or systems.
 - Prevents impersonation attacks.
 - Uses passwords, OTP, biometrics.
-

◆ 2. Access Control Service

- Controls who can access what resources.
 - Enforces authorization policies.
 - Example: Role-based access (admin/user).
-

◆ 3. Data Confidentiality Service

- Protects data from unauthorized access.
 - Uses encryption techniques.
-

◆ 4. Data Integrity Service

- Ensures data is not modified during transmission or storage.
 - Uses hashing and MAC.
-

◆ 5. Non-repudiation Service

- Provides proof of data origin and delivery.

- Prevents denial of communication.

◆ 6. Availability Service

- Ensures system is always available to users.
 - Prevents DoS and system failures.
-

◆ 3. Relationship Between Principles and Services

- Security principles = Goals (what we want to achieve)
- Security services = Methods (how we achieve them)

👉 Example:

- Confidentiality (principle) → Encryption (service)
- Integrity (principle) → Hashing (service)

Q16. Write comparison between symmetric and asymmetric key cryptography

ANS:

Feature	Symmetric Key Cryptography	Asymmetric Key Cryptography
Number of Keys	Single key	Two keys (public & private)
Key Usage	Same key for encryption & decryption	Different keys for encryption & decryption
Speed	Very fast	Slower
Security	Moderate	High security
Key Distribution	Difficult and risky	Easy (public key sharing)

Feature	Symmetric Key Cryptography	Asymmetric Key Cryptography
Computational Complexity	Low	High
Algorithm Examples	AES, DES, 3DES	RSA, Diffie-Hellman, ECC
Data Size Handling	Suitable for large data	Suitable for small data
Performance	High performance	Lower performance
Usage	Bulk encryption	Key exchange, authentication, digital signature

UNIT 5

Q1. Explain PKIX Model.

ANS:

PKIX Model (Public Key Infrastructure using X.509) – Explanation

◇ **1. Introduction**

- **PKIX (Public Key Infrastructure using X.509) is a standardized model used to manage public key certificates and digital identities.**
 - **It is based on the X.509 certificate standard and is used in secure communication systems like HTTPS, email security, VPNs, etc.**
 - **PKIX is defined by the IETF (Internet Engineering Task Force).**
-

◇ **2. Definition**

- **The PKIX model is a framework that defines how public key certificates are created, distributed, validated, and revoked using trusted authorities.**
-

◇ **3. Purpose of PKIX Model**

- **To ensure secure authentication of users and systems**
 - **To manage public key certificates**
 - **To provide trust between communicating parties**
 - **To support digital signatures and encryption**
-

◇ **4. Components of PKIX Model**

◇ **1. End Entity**

- **The user or system (client/server) that uses certificates**

- **Example: browser, email user**
-

◇ 2. Certificate Authority (CA)

- **A trusted third party that issues digital certificates**
 - **Verifies identity before issuing certificate**
 - **Acts like a “digital passport authority”**
-

◇ 3. Registration Authority (RA)

- **Works as a middle layer between user and CA**
 - **Performs identity verification before certificate approval**
-

◇ 4. Certificate Repository

- **Stores issued certificates**
 - **Publicly accessible database or directory**
-

◇ 5. Certificate Revocation System

- **Maintains list of invalid or revoked certificates**
 - **Uses:**
 - **CRL (Certificate Revocation List)**
 - **OCSP (Online Certificate Status Protocol)**
-

◇ 5. PKIX Architecture Diagram

Pending dia blurr

◇ 6. Working of PKIX Model

◇ Step 1: Request

- User requests a digital certificate.

◇ Step 2: Verification

- RA verifies user identity.

◇ Step 3: Certificate Issuance

- CA issues X.509 digital certificate containing:
 - Public key
 - Identity details
 - Digital signature of CA

◇ Step 4: Distribution

- Certificate is stored in repository and shared.

◇ Step 5: Verification

- Receiver verifies certificate using CA's public key.

◇ Step 6: Revocation (if needed)

- If compromised, certificate is added to CRL.

◇ 7. Features of PKIX Model

- Based on X.509 certificate standard
- Provides trust hierarchy (CA-based trust)
- Supports certificate validation and revocation
- Ensures secure communication over insecure networks

◇ 8. Advantages

- Strong authentication system

- **Widely used in internet security**
 - **Provides scalable trust model**
 - **Supports encryption and digital signatures**
-

◇ 9. Disadvantages

- **Complex infrastructure**
 - **Dependence on trusted CA**
 - **Certificate management overhead**
-

◇ 10. Applications

- **HTTPS websites (SSL/TLS)**
- **Secure email (S/MIME)**
- **VPN authentication**
- **E-commerce transactions**

Q2. Explain Advanced Encryption Standard with diagram

ANS:

Advanced Encryption Standard (AES)

◇ 1. Introduction

- **AES (Advanced Encryption Standard) is a symmetric key block cipher algorithm used to secure sensitive data.**
- **It was established by the NIST (National Institute of Standards and Technology) in 2001.**
- **AES replaced DES due to its higher security and efficiency.**

- It is widely used in Wi-Fi security, banking systems, SSL/TLS, and file encryption.
-

◇ 2. Key Features of AES

- Symmetric key algorithm
 - Block size = 128 bits fixed
 - Key sizes = 128, 192, or 256 bits
 - Strong resistance to cryptanalysis
 - Works on substitution–permutation network (SPN) structure
-

◇ 3. AES Structure

- AES operates on a 4×4 byte matrix called State Array
 - Data is processed in multiple rounds:
 - 10 rounds (AES-128)
 - 12 rounds (AES-192)
 - 14 rounds (AES-256)
-

◇ 4. AES Encryption Process

AES encryption consists of the following steps:

◇ Step 1: Key Expansion

- Original key is expanded into multiple round keys using a key schedule.
-

◇ Step 2: Initial Round

- AddRoundKey operation

- Plaintext is XORed with the first round key.
-

◇ Step 3: Main Rounds (Repeated)

Each round includes:

1. SubBytes

- Byte substitution using S-box

2. ShiftRows

- Rows are cyclically shifted

3. MixColumns

- Columns are mixed using mathematical transformation

4. AddRoundKey

- XOR with round key
-

◇ Step 4: Final Round

- Same as main round but without MixColumns



◇ 6. AES Decryption Process

- Reverse operations are applied:

- Inverse SubBytes
 - Inverse ShiftRows
 - Inverse MixColumns
 - AddRoundKey
-

◇ 7. AES Round Structure (Simple View)

State → SubBytes → ShiftRows → MixColumns → AddRoundKey

◇ 8. Working Principle

- AES is based on Substitution–Permutation Network (SPN)
 - It ensures:
 - Confusion (SubBytes)
 - Diffusion (ShiftRows + MixColumns)
 - These properties make AES highly secure against attacks
-

◇ 9. Advantages of AES

- Very high security level
 - Fast in both software and hardware
 - Resistant to known attacks
 - Efficient for large data encryption
 - Widely accepted global standard
-

◇ 10. Disadvantages of AES

- Complex implementation compared to DES
- Requires more computational power than simpler ciphers

- Key management still important
-

◇ 11. Applications

- HTTPS (secure websites)
- Wi-Fi security (WPA2/WPA3)
- Banking transactions
- Cloud storage encryption
- Government classified data protection

Q3. Explain following terms i) iHash Function ii) Digital Signature iii) Digital Certificate

What are the applications OF it

ANS: 1) Hash Function (Detailed Theory)

 **Definition:**

A hash function is a cryptographic algorithm that takes an input (message of any size) and produces a fixed-size output called hash value or message digest.

It is mainly used to ensure data integrity.

 **Working Principle:**

- Input message is processed through a mathematical function.
 - Output is a unique fixed-length value.
 - Same input always gives same output.
-

 **Characteristics:**

- ✓ One-way function (cannot reverse)

- ✓ Deterministic (same input → same output)
 - ✓ Fast computation
 - ✓ Avalanche effect (small change → big hash change)
 - ✓ Collision resistant (very difficult to get same hash for two inputs)
-

Example Algorithms:

- MD5 (old, not secure now)
 - SHA-1 (weak)
 - SHA-256 (secure and widely used)
-

Uses:

- Password storage
 - File integrity checking
 - Digital signatures
 - Blockchain
-

☒ 2) Digital Signature (Detailed Theory)

Definition:

A Digital Signature is a cryptographic technique used to verify:

- ✓ Authenticity (who sent the message)
 - ✓ Integrity (message not changed)
 - ✓ Non-repudiation (sender cannot deny sending)
-

Working Process:

✓ Step 1: Sender side

- Message is passed through hash function → Message digest
- Digest is encrypted using sender's private key
- This encrypted digest = Digital Signature

✓ Step 2: Receiver side

- Digital signature is decrypted using sender's public key
- Receiver generates hash of received message
- Both hashes are compared

Diagram:

SENDER SIDE:

Message → Hash Function → Digest → Encrypt (Private Key) → Digital Signature

RECEIVER SIDE:

Digital Signature → Decrypt (Public Key) → Digest
Message → Hash Function → New Digest

Compare both digests

Properties:

- ✓ Authentication
- ✓ Integrity
- ✓ Non-repudiation
- ✓ Security against tampering

 Applications:

- Online banking transactions
- E-mail security
- Software downloads
- E-Governance systems

☒ **3) Digital Certificate (Detailed Theory)**

 Definition:

A Digital Certificate is an electronic document issued by a trusted authority called Certificate Authority (CA) that binds:

- A public key
- With identity of a person/organization

It is used in Public Key Infrastructure (PKI).

 Purpose:

- To verify identity of user or website
- To ensure secure communication

 Contents of Digital Certificate:

- Owner name (user/organization)
- Public key
- Issuer (CA name)
- Validity period
- Serial number

- **Digital signature of CA**
-

 **Diagram:**

User Identity + Public Key



Certificate Authority (CA)



Digital Certificate Issued

 **Working:**

1. **User generates key pair (Public + Private key)**
 2. **Public key is sent to CA**
 3. **CA verifies identity**
 4. **CA signs and issues certificate**
 5. **Others trust the certificate via CA**
-

 **Example:**

- **HTTPS websites (SSL/TLS certificates)**
 - **Banking websites (padlock icon in browser)**
-

 **Applications (Detailed)**

 **1. Secure Communication**

- **Used in HTTPS websites**
 - **Ensures encrypted communication**
-

2. E-Mail Security

- Digital signature ensures sender authenticity
 - Prevents fake/spam emails
-

3. Online Banking

- Protects transactions
 - Ensures user identity
-

4. Software Security

- Verifies software authenticity
 - Prevents malware injection
-

5. Blockchain & Cryptocurrency

- Hashing ensures data integrity
 - Digital signatures verify transactions
-

6. E-Governance

- Aadhaar, digital ID systems
- Secure government document verification

Q4. Explain the working of DES algorithm with suitable example

ANS: Working of DES Algorithm with Suitable Example

1) Introduction

DES (Data Encryption Standard) is a symmetric key block cipher used to encrypt 64-bit data blocks using a 56-bit secret key. It uses a Feistel structure with 16 rounds of processing.

2) Basic Idea of DES Working

- Input: 64-bit plaintext
 - Output: 64-bit ciphertext
 - Process: Substitution + Permutation + Key mixing
 - Structure: Feistel Network (same structure for encryption & decryption)
-

3) DES Encryption Steps

◇ Step 1: Initial Permutation (IP)

- The 64-bit plaintext is rearranged using a fixed table.
 - This does NOT add security but improves diffusion.
-

◇ Step 2: Split into Two Halves

After IP:

- Left half = L0 (32 bits)
 - Right half = R0 (32 bits)
-

◇ Step 3: 16 Feistel Rounds

Each round uses a subkey K_i (48-bit).

🔗 Round formula:

$$L_i = R_{i-1}, R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

◇ Step 4: F Function (Core of DES)

Steps inside F:

1. Expansion (32 → 48 bits)
 2. XOR with round key K_i
 3. S-box substitution (48 → 32 bits)
 4. P-box permutation
-

◇ Step 5: Swap

After 16 rounds:

- Swap left and right halves
-

◇ Step 6: Final Permutation (FP)

- Reverse of initial permutation
 - Produces ciphertext
-

📊 DES Working Diagram

Plaintext (64-bit)



Initial Permutation (IP)



Split: L0 (32) | R0 (32)



16 ROUNDS



Swap (L16, R16)



Final Permutation (FP)



Ciphertext (64-bit)



Key Generation (Important)

- **Input key: 64-bit**
- **Effective key: 56-bit (8 parity bits removed)**

Steps:

1. **PC-1 permutation → split into C0 & D0 (28 bits each)**
 2. **Left shifts in each round**
 3. **PC-2 permutation → generate 48-bit subkey K_i**
 4. **Total subkeys = 16**
-



4) Suitable Example (Simple Conceptual)

◇ Given:

- **Plaintext: 0123456789ABCDEF (hex)**
- **Key: 133457799BBCDFF1**

Step-wise Working:

✓ Step 1: Initial Permutation

Plaintext is rearranged bitwise.

✓ Step 2: Split

- L_0 = first 32 bits
 - R_0 = last 32 bits
-

✓ Step 3: First Round Example

$F(R_0, K_1)$:

- Expand $R_0 \rightarrow 48$ bits
- XOR with K_1
- Apply S-box $\rightarrow 32$ bits
- Apply P-box

Then:

$$L_1 = R_0, R_1 = L_0 \oplus F(R_0, K_1)$$

✓ Step 4: Repeat for 16 Rounds

- Each round uses different subkey (K_1 to K_{16})
 - Data becomes more scrambled after each round
-

✓ Step 5: Final Step

- Swap halves
- Apply Final Permutation
- Output ciphertext

Final Output Example (Conceptual)

After processing:

Plaintext: 0123456789ABCDEF

Ciphertext: 85E813540F0AB405

5) Decryption Process

- Same steps as encryption
 - Only difference:
 -  Subkeys used in reverse order: K16 → K1
-

Key Points of DES Working

- ✓ 16 Feistel rounds
 - ✓ Substitution + permutation + XOR
 - ✓ Strong confusion and diffusion
 - ✓ Same structure for encryption & decryption
-

Limitations

- Weak due to 56-bit key
- Vulnerable to brute force attacks
- Replaced by AES

Q5. Define symmetric key encryption? Explain DES in detail.

ANS: Symmetric Key Encryption – Definition

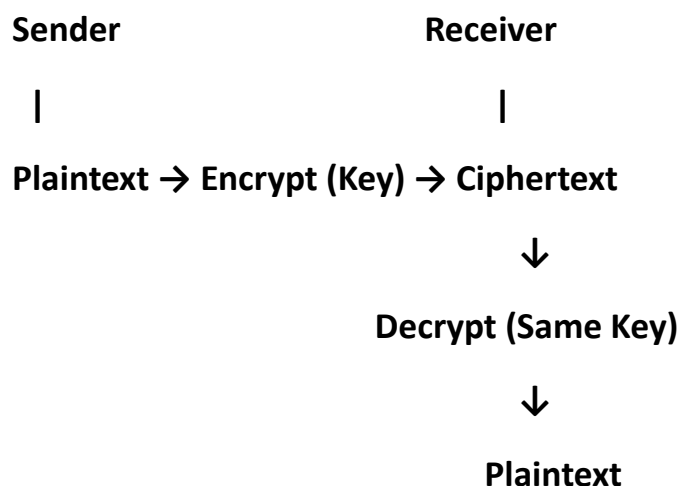
 **Definition:**

Symmetric key encryption is a cryptographic technique in which the same secret key is used for both encryption and decryption of data.

 **Key Points:**

- ✓ One common key is shared between sender and receiver
 - ✓ Faster than asymmetric encryption
 - ✓ Requires secure key distribution
 - ✓ Also called Secret Key Cryptography
-

 **Diagram:**



 **Examples:**

- DES
- AES
- Blowfish

2) DES (Data Encryption Standard) – Detailed Explanation

Definition:

DES is a symmetric key block cipher algorithm that encrypts data in 64-bit blocks using a 56-bit secret key. It uses a Feistel network with 16 rounds of processing.

Basic Features:

- Block size: 64 bits
 - Key size: 56 bits (effective)
 - Rounds: 16
 - Structure: Feistel cipher
 - Type: Symmetric encryption
-

DES Working Steps

◇ Step 1: Initial Permutation (IP)

- Rearranges the 64-bit plaintext using a fixed table
 - No security added, only diffusion
-

◇ Step 2: Split Data

After IP:

- Left half = L0 (32 bits)
- Right half = R0 (32 bits)

◇ Step 3: 16 Feistel Rounds

Each round uses a subkey K_i (48-bit).

✂ Round Formula:

$$L_i = R_{i-1}, R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

◇ Step 4: F Function (Core of DES)

The function F includes:

✓ 1. Expansion

- 32-bit → 48-bit

✓ 2. XOR with Key

- XOR with 48-bit subkey

✓ 3. S-box Substitution

- Converts 48-bit → 32-bit
- Provides non-linearity

✓ 4. P-box Permutation

- Rearranges bits for diffusion
-

 F Function Flow:

R (32-bit)

↓ Expansion

48-bit

↓ XOR with K_i

48-bit

↓ S-box

32-bit

↓ P-box

$F(R, K_i)$

◇ Step 5: Swap

After 16 rounds:

- Swap L16 and R16
-

◇ Step 6: Final Permutation (FP)

- Inverse of Initial Permutation
 - Produces final ciphertext
-

🔄 DES Structure Diagram:

Plaintext (64-bit)

↓

Initial Permutation (IP)

↓

Split (L0 | R0)

↓

16 Feistel Rounds

↓

Swap

↓

Final Permutation (FP)



Ciphertext (64-bit)

Key Generation Process:

1. 64-bit key input
 2. Remove parity bits → 56-bit key
 3. PC-1 permutation → split into C0 & D0 (28 bits each)
 4. Left shifts in each round
 5. PC-2 permutation → generate 48-bit subkeys
 6. Total subkeys = 16 (K1 to K16)
-

Decryption:

- Same process as encryption
 - Only difference:  Subkeys used in reverse order (K16 → K1)
-

Advantages of DES:

- Fast encryption (hardware friendly)
 - Strong Feistel structure
 - Easy implementation
-

Disadvantages of DES:

- Small key size (56-bit → weak today)
 - Vulnerable to brute-force attack
 - Replaced by AES
-

Applications:

- ATM systems (old systems)
- Secure file encryption
- Legacy banking systems
- Network security (historical use)

Q6. Write a note on RSA algorithm WITH EX

ANS:

RSA Algorithm – Complete Answer

1) Introduction

RSA (Rivest–Shamir–Adleman) is a public key (asymmetric) cryptographic algorithm used for secure data transmission. It uses two different keys, one for encryption and one for decryption.

- Public key is used for encryption
 - Private key is used for decryption
 - It is based on the difficulty of factoring large prime numbers
-

2) Key Idea

RSA works on the concept:

- Choose two large prime numbers
 - Multiply them to get a modulus
 - Use modular exponentiation for encryption and decryption
-

3) RSA Key Generation Steps

Step 1: Choose two prime numbers

Let

$$p = 3$$

$$q = 5$$

Step 2: Compute n

$$n = p \times q$$

$$n = 3 \times 5 = 15$$

Step 3: Compute Euler's Totient Function

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (3 - 1)(5 - 1) = 2 \times 4 = 8$$

Step 4: Choose public key e

Choose e such that:

- $1 < e < \phi(n)$
- $\gcd(e, \phi(n)) = 1$

Take $e = 3$

Public key = $(e, n) = (3, 15)$

Step 5: Compute private key d

Find d such that:

$$(e \times d) \bmod \phi(n) = 1$$

$$(3 \times d) \bmod 8 = 1$$

$$d = 3$$

Private key = $(d, n) = (3, 15)$

4) Encryption Process

Formula:

$$C = M^e \bmod n$$

Example:

Let message $M = 2$

$$C = 2^3 \bmod 15$$

$$C = 8 \bmod 15 = 8$$

Ciphertext = 8

5) Decryption Process

Formula:

$$M = C^d \bmod n$$

$$M = 8^3 \bmod 15$$

$$8^3 = 512$$

$$512 \bmod 15 = 2$$

Original message = 2

6) Working Diagram

Sender Side:

Plaintext (M)



Encryption using Public Key (e, n)



Ciphertext (C)

Receiver Side:

Ciphertext (C)



Decryption using Private Key (d, n)



Plaintext (M)

7) Properties of RSA

- **Uses two different keys (public and private)**
 - **Secure due to difficulty of factoring large numbers**
 - **Supports encryption and digital signatures**
 - **Widely used in secure communication**
-

8) Applications of RSA

- **Secure web communication (HTTPS/SSL)**
 - **Digital signatures**
 - **Email security**
 - **Online banking systems**
 - **Software authentication**
-

9) Advantages

- **Strong security**
- **No need to share private key**
- **Supports authentication and integrity**

10) Disadvantages

- Slow compared to symmetric algorithms
- Requires large key size for strong security
- Computationally expensive

Q7. What is private key? Explain Diffie-Hellman key exchange algorithm

ANS:

Private Key – Detailed Theory (Simple Language)

 **Definition:**

A private key is a secret key used in asymmetric cryptography that is known only to the owner and is used for decrypting data and creating digital signatures.

 **Explanation:**

In public key cryptography, every user has a key pair:

- Public key → shared with everyone
- Private key → kept secret

The private key works like a master key that can unlock the encrypted message sent using the public key.

 **Working:**

1. Sender encrypts data using receiver's public key
2. Encrypted data is sent over network
3. Receiver uses private key to decrypt it
4. Only receiver can read the original message

Importance:

- Ensures data confidentiality
- Used for secure communication
- Used in digital signatures for authentication

Properties:

- Always secret
- Cannot be shared
- Mathematically linked with public key
- Very important for security

Example:

- In RSA:
 - Public key → encrypt message
 - Private key → decrypt message

Simple Analogy:

- Public key = lock (anyone can lock)
- Private key = unique key (only owner can open)

Conclusion:

Private key is the most important part of asymmetric cryptography and is used to ensure secure communication and authentication.

2) Diffie–Hellman Key Exchange – Detailed Theory (Simple)

Definition:

Diffie–Hellman Key Exchange is a cryptographic method used to securely generate a shared secret key between two users over an insecure network without sending the key directly.

Main Idea:

Two users (Alice and Bob) want to communicate securely.

Instead of sharing a secret key directly, they create the same secret key using mathematical calculations.

Working Steps:

Step 1: Public Values

Both users agree on:

- A large prime number (q)
- A base number (α)

These are public and known to everyone.

Step 2: Private Values

- Alice selects a secret number (A)
- Bob selects a secret number (B)

These are never shared.

Step 3: Generate Public Values

Alice computes:

$$Y_A = \alpha^A \bmod q$$

Bob computes:

$$Y_B = \alpha^B \bmod q$$

They exchange Y_A and Y_B .

Step 4: Generate Shared Secret Key

Alice computes:

$$K = Y_B^A \bmod q$$

Bob computes:

$$K = Y_A^B \bmod q$$

Both get the same key:

$$K = \alpha^{AB} \bmod q$$

Why it is Secure:

- Secret numbers A and B are never sent
 - Only calculated values are exchanged
 - Hard to find private values from public values (mathematical difficulty)
-

Applications:

- HTTPS (secure websites)
 - VPN connections
 - Secure messaging apps (WhatsApp, Signal)
 - Network security protocols
-

Advantages:

- No need to send secret key

- Secure over public network
 - Easy to implement
-

✗ Disadvantages:

- No authentication (can be attacked by Man-in-the-Middle)
- Needs extra security methods
- Slow for large computations

Q9. Explain following terms: i) ii) Groups Rings iii) Prime Numbers

ANS:

1) Groups

A group is a mathematical structure consisting of a set of elements together with an operation that combines any two elements to form another element of the same set.

A set G with operation $(*)$ is called a group if it satisfies the following properties:

1. Closure Property:

If a and b are in G , then

$a * b$ is also in G .

2. Associative Property:

$$(a * b) * c = a * (b * c)$$

3. Identity Element:

There exists an element e in G such that

$$a * e = e * a = a$$

4. Inverse Element:

For every element a in G , there exists an element a^{-1} such that $a * a^{-1} = e$

Example:

Set of integers under addition $(\mathbb{Z}, +)$ is a group:

- Closure: $a + b$ is an integer
 - Identity: 0
 - Inverse: $-a$ exists
-

Importance in Cryptography:

Groups are used in:

- Modular arithmetic operations
 - RSA algorithm
 - Diffie-Hellman key exchange
-

2) Rings

A ring is a mathematical structure consisting of a set equipped with two operations: addition and multiplication.

A set R with operations $(+, \times)$ is called a ring if it satisfies:

1. $(R, +)$ forms a group:

- Addition is closed
- Has identity element (0)
- Every element has inverse

2. Closure under multiplication:

If a and b are in R , then

$a \times b$ is also in R

3. Associative property of multiplication:

$$(a \times b) \times c = a \times (b \times c)$$

4. Distributive laws:

$$a \times (b + c) = a \times b + a \times c$$

$$(b + c) \times a = b \times a + c \times a$$

Example:

Set of integers ($\mathbb{Z}$) under addition and multiplication forms a ring.

Importance in Cryptography:

- Used in RSA algorithm
 - Helps in modular arithmetic calculations
 - Important for designing encryption systems
-

3) Prime Numbers

A prime number is a natural number greater than 1 that has only two factors:

- 1
 - itself
-

Examples:

2, 3, 5, 7, 11, 13, 17, 19, etc.

Properties:

- **Only divisible by 1 and itself**
 - **2 is the only even prime number**
 - **Infinite prime numbers exist**
 - **Fundamental in number theory**
-

Importance in Cryptography:

1. RSA Algorithm:

- **Uses two large prime numbers (p and q)**
 - **Security depends on difficulty of factoring their product**
-

2. Key Generation:

- **Used to generate secure keys**
- **Ensures strong encryption**

Why Prime Numbers are Important:

- **Hard to factor large numbers**
- **Provides strong security**
- **Backbone of public key cryptography**

Q10. Define and list various computer network security mechanisms Also write short notes on the following terms i) Encryption ii) Decryption

ANS:

1) Computer Network Security Mechanisms

 **Definition:**

Network security mechanisms are the techniques, tools, and processes used to protect computer networks and data from unauthorized access, attacks, and misuse.

They ensure confidentiality, integrity, authentication, and availability (CIA properties) of data.

 **Types / List of Security Mechanisms:**

1. Encryption Mechanism

- **Converts readable data (plaintext) into unreadable form (ciphertext)**
 - **Ensures confidentiality**
-

2. Authentication Mechanism

- **Verifies identity of user or system**
 - **Example: passwords, OTP, biometrics**
-

3. Access Control Mechanism

- **Restricts access to authorized users only**
 - **Example: user login permissions**
-

4. Digital Signature Mechanism

- **Ensures authentication, integrity, and non-repudiation**

- Used in emails and online transactions
-

5. Firewalls

- Acts as a barrier between trusted and untrusted networks
 - Filters incoming and outgoing traffic
-

6. Intrusion Detection System (IDS)

- Detects unauthorized access or attacks
 - Monitors network traffic
-

7. Hashing Mechanism

- Ensures data integrity using hash functions
 - Detects changes in data
-

8. VPN (Virtual Private Network)

- Provides secure communication over public networks
- Encrypts entire communication channel

2) Short Notes

i) Encryption

 Definition:

Encryption is the process of converting plaintext (readable data) into ciphertext (unreadable form) using an algorithm and a secret key.

 Explanation:

Encryption is used to protect data during transmission over insecure networks. Even if data is intercepted, it cannot be understood without the key.

 **Process:**

Plaintext + Encryption Algorithm + Key = Ciphertext

 **Characteristics:**

- Ensures confidentiality
 - Prevents unauthorized access
 - Uses cryptographic algorithms
-

 **Types:**

- Symmetric encryption (same key for encryption and decryption)
 - Asymmetric encryption (public and private keys)
-

 **Example:**

HELLO → encrypted → X9@B7

 **Applications:**

- Online banking
 - Secure messaging
 - E-commerce transactions
-

ii) Decryption

 **Definition:**

Decryption is the process of converting ciphertext back into original plaintext using a decryption algorithm and a key.

Explanation:

It is the reverse process of encryption. Only authorized users with the correct key can perform decryption.

Process:

Ciphertext + Decryption Algorithm + Key = Plaintext

Characteristics:

- **Restores original data**
 - **Ensures authorized access**
 - **Requires correct key**
-

Example:

X9@B7 → decrypted → HELLO

Types:

- **Symmetric decryption (same key used)**
 - **Asymmetric decryption (private key used)**
-

Applications:

- **Reading secure messages**
- **Accessing encrypted files**

- Secure communication systems

Q11. Explain the working of DES with suitable example

ANS:

Working of DES Algorithm with Suitable Example

1) Introduction

DES (Data Encryption Standard) is a symmetric key block cipher algorithm used to encrypt 64-bit data blocks using a 56-bit secret key. It works on substitution and permutation and uses 16 rounds (Feistel structure).

2) Basic Idea

- Input: 64-bit plaintext
 - Output: 64-bit ciphertext
 - Key: 56-bit effective key
 - Process: 16 rounds of mixing data and key
-

3) Steps in DES Working

Step 1: Initial Permutation (IP)

- The 64-bit plaintext is rearranged using a fixed table
 - This improves diffusion (spreading of bits)
-

Step 2: Split into Two Halves

After IP:

- Left half = L0 (32 bits)

- Right half = R0 (32 bits)
-

Step 3: 16 Feistel Rounds

Each round uses a subkey K_i (48-bit).

Round operation:

$$L_i = R_{i-1} \text{quad}, \text{quad} R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Step 4: F Function (Core of DES)

F function has 4 steps:

1. Expansion:

- R (32-bit) $\rightarrow$ 48-bit

2. XOR with Key:

- 48-bit data XOR 48-bit subkey K_i

3. Substitution (S-box):

- Converts 48-bit $\rightarrow$ 32-bit
- Provides confusion (security)

4. Permutation (P-box):

- Rearranges bits for diffusion
-

Step 5: Swap

After 16 rounds:

- Swap L16 and R16
-

Step 6: Final Permutation (FP)

- Inverse of Initial Permutation

- Produces final ciphertext
-

4) DES Working Diagram

Plaintext (64-bit)



Initial Permutation (IP)



Split → L0 (32) | R0 (32)



16 Feistel Rounds



Swap (L16, R16)



Final Permutation (FP)



Ciphertext (64-bit)Ciphertext (64-bit)

5) Key Generation (Brief)

- 64-bit key input
 - 8 parity bits removed → 56-bit key
 - Split into two halves
 - Left shifts each round
 - 16 subkeys (K1 to K16) generated
-

6) Suitable Example (Simple Conceptual)

Given:

- Plaintext = 0123456789ABCDEF (hex)
 - Key = 133457799BBCDFF1
-

Step 1: Initial Permutation

- Bits are rearranged
-

Step 2: Split

- L0 = first 32 bits
 - R0 = last 32 bits
-

Step 3: First Round Example

F(R0, K1):

- Expand R0
- XOR with K1
- Apply S-box
- Apply P-box

Then:

$$L_1 = R_0 \text{quad}, \text{quad} R_1 = L_0 \oplus F(R_0, K_1)$$

Step 4: Repeat

- Same process repeated for 16 rounds using K1 to K16
-

Step 5: Final Output

After all rounds and final permutation:

Example result:

- Ciphertext = 85E813540F0AB405 (hex)
-

7) Decryption

- Same process as encryption
 - Only difference:
 - 👉 Subkeys used in reverse order (K16 → K1)
-

8) Key Points of DES

- Uses 16 rounds
- Feistel structure
- Substitution + permutation
- Same structure for encryption and decryption

Q12. Describe how digital signature standard scheme generate and verify the digital signature

ANS:

Digital Signature Standard (DSS) – Generation and Verification

📌 **Definition:**

Digital Signature Standard (DSS) is a cryptographic method used to create and verify digital signatures. It ensures authentication, integrity, and non-repudiation of a message using public key cryptography.

1) Generation of Digital Signature

📌 **Steps:**

Step 1: Key Generation

- Sender generates a private key and a public key
 - Private key is kept secret
 - Public key is shared with others
-

Step 2: Hashing the Message

- The message is passed through a hash function
 - This produces a fixed-size value called message digest
-

Step 3: Signature Creation

- The sender uses:
 - Message digest
 - Private key
 - A digital signature is created using a cryptographic algorithm
-

 **Result:**

Message + Digital Signature is sent to the receiver

 **Diagram (Generation):**

Message → Hash Function → Message Digest



Private Key



Digital Signature

2) Verification of Digital Signature

 Steps:

Step 1:

Receiver gets:

- Message
 - Digital signature
-

Step 2:

- Receiver generates hash of received message
-

Step 3:

- Receiver uses sender's public key to verify the signature
 - This produces original message digest
-

Step 4:

- Both digests are compared:
 - If same → signature is valid
 - If different → signature is invalid

 Diagram (Verification):

Message + Signature



Hash Function



Message Digest



Compare with decrypted signature



Valid / Invalid

UNIT 6

Q1. What is the difference between hardware attack and software attack?

Write a note on MIM attack

ANS:

No.	Hardware Attack	Software Attack
1	It is an attack on physical components of a system	It is an attack on programs or applications
2	Targets hardware devices like CPU, RAM, disk, keyboard	Targets software like OS, applications, browsers
3	Requires physical access in most cases	Can be done remotely over a network
4	Involves physical damage or manipulation	Involves logical manipulation of code or data
5	Harder to detect because it is physical	Easier to detect using antivirus tools
6	Example: hardware keylogger, device tampering	Example: virus, worm, ransomware
7	Can permanently damage system hardware	Usually affects data or software functionality
8	Prevention includes physical security (locks, CCTV)	Prevention includes antivirus, firewall, updates
9	Less common but more dangerous physically	More common in cyber world
10	Requires direct interaction with system	Does not require physical presence

Note on Man-in-the-Middle (MITM) Attack

Definition:

A Man-in-the-Middle (MITM) attack is a type of network security attack in which an attacker secretly intercepts and possibly alters communication between two legitimate users without their knowledge.

Explanation:

In a MITM attack, the attacker positions himself between the sender and receiver. Both users believe they are communicating directly with each other, but in reality, all messages pass through the attacker.

The attacker can:

- Read the data (eavesdropping)
 - Modify the data (data tampering)
 - Inject fake messages
-

Working Diagram:

Sender $\leftrightarrow$ Attacker $\leftrightarrow$ Receiver

(Intercepts and modifies communication)

Working Steps:

1. Sender sends data to receiver
 2. Attacker secretly intercepts the data
 3. Attacker reads or modifies the message
 4. Modified or original message is forwarded to receiver
 5. Receiver is unaware of the attack
-

Types of MITM Attacks:

- **IP spoofing**
 - **Session hijacking**
 - **Email hijacking**
 - **Wi-Fi eavesdropping**
 - **DNS spoofing**
-

Effects of MITM Attack:

- **Theft of sensitive information (passwords, bank details)**
 - **Data manipulation**
 - **Identity theft**
 - **Financial loss**
 - **Loss of privacy**
-

Prevention Methods:

- **Use HTTPS (SSL/TLS encryption)**
 - **Use strong encryption techniques**
 - **Avoid public Wi-Fi for sensitive transactions**
 - **Use VPN (Virtual Private Network)**
 - **Use multi-factor authentication**
 - **Verify digital certificates**
-

Example:

If a user logs into a banking website on an insecure network, an attacker can intercept the username and password and gain unauthorized access to the account.

Q2. Define Vulnerability. Explain types of vulnerabilities in detail

ANS:  Vulnerability

 **Definition:**

A vulnerability is a weakness, flaw, or gap in a computer system, network, software, hardware, or human process that can be exploited by an attacker to perform unauthorized actions such as accessing data, modifying information, or disrupting services.

 **Explanation:**

- Vulnerabilities are not attacks themselves, but they enable attacks.
 - They may arise due to poor design, coding errors, improper configuration, or lack of security awareness.
 - Attackers identify these weaknesses and exploit them using different techniques.
 - If vulnerabilities are not fixed, they can lead to serious security breaches.
-

 **Example:**

- Weak password like “1234”
 - Unpatched operating system
 - Open network ports
-

 **Types of Vulnerabilities (Detailed Explanation)**

1) Software Vulnerabilities

Definition:

Weaknesses in software programs or operating systems caused by coding errors, bugs, or poor design.

Explanation:

Software vulnerabilities occur when developers fail to handle input properly or do not secure the code. Attackers exploit these flaws to execute malicious code or gain unauthorized access.

Examples:

- Buffer overflow
 - SQL Injection
 - Cross-Site Scripting (XSS)
 - Unpatched OS bugs
-

Impact:

- Data leakage
 - Unauthorized access
 - System crashes
-

Prevention:

- Regular software updates
- Secure coding practices
- Input validation

2) Hardware Vulnerabilities

Definition:

Weaknesses in physical components of a system.

Explanation:

These occur due to design flaws or physical access to hardware. Attackers can manipulate or damage hardware to extract or alter data.

Examples:

- Hardware keyloggers
 - Faulty chips
 - Side-channel attacks
-

Impact:

- Permanent damage
 - Data theft
 - System malfunction
-

Prevention:

- Physical security measures
 - Secure hardware design
 - Restricted access
-

3) Network Vulnerabilities

Definition:

Weaknesses in network infrastructure or communication systems.

Explanation:

These vulnerabilities arise due to improper network configuration or lack of encryption, allowing attackers to intercept or modify data.

Examples:

- **Open ports**
 - **Weak Wi-Fi security**
 - **Unsecured protocols (HTTP)**
 - **Man-in-the-Middle attacks**
-

Impact:

- **Data interception**
 - **Unauthorized access**
 - **Network disruption**
-

Prevention:

- **Use encryption (HTTPS, VPN)**
- **Secure network configuration**
- **Firewalls and IDS**

4) Configuration Vulnerabilities

Definition:

Weaknesses caused by incorrect system or software settings.

Explanation:

Even secure systems can become vulnerable if they are not configured properly. Default settings and unnecessary services increase risk.

Examples:

- Default usernames/passwords
 - Misconfigured firewall
 - Unused open services
-

Impact:

- Easy exploitation
 - Unauthorized access
-

Prevention:

- Change default settings
 - Disable unused services
 - Proper system configuration
-

5) Human (Social Engineering) Vulnerabilities

Definition:

Weaknesses caused by human errors or behavior.

 Explanation:

Attackers exploit human psychology rather than technical flaws. Lack of awareness leads to security breaches.

 Examples:

- **Phishing emails**
 - **Sharing passwords**
 - **Clicking malicious links**
 - **Using weak passwords**
-

 Impact:

- **Identity theft**
 - **Financial fraud**
 - **Data loss**
-

 Prevention:

- **User awareness training**
- **Strong passwords**
- **Avoid suspicious links**

Q3. Define cyber security and explain the different layers of security.

ANS: Cyber Security

 **Definition:**

Cyber security is the discipline of protecting computer systems, networks, devices, and data from unauthorized access, cyber attacks, damage, or theft using technologies, processes, and policies.

 **Detailed Explanation:**

- With the rapid growth of the internet, cyber threats have increased significantly.
 - Cyber security ensures that systems remain secure, reliable, and trustworthy.
 - It protects against threats like malware, phishing, hacking, denial of service attacks, and data breaches.
 - It focuses on maintaining the CIA triad:
 - Confidentiality – data is accessible only to authorized users
 - Integrity – data remains accurate and unaltered
 - Availability – data and services are available when needed
-

 **Goals of Cyber Security:**

- Prevent unauthorized access
 - Protect sensitive data
 - Ensure safe communication
 - Maintain system performance and availability
 - Detect and respond to cyber threats
-

Layers of Security (Defense in Depth Concept)

Cyber security uses multiple layers of protection. If one layer fails, other layers continue to provide security.

1) Physical Security Layer

Explanation:

This layer protects the physical infrastructure such as servers, computers, and networking devices.

Measures:

- Locked server rooms
- CCTV surveillance
- Biometric access systems
- Security personnel

Importance:

Prevents unauthorized physical access and hardware tampering.

2) Perimeter / Network Security Layer

Explanation:

This layer protects the network boundaries and controls incoming and outgoing traffic.

Measures:

- Firewalls
- Intrusion Detection/Prevention Systems (IDS/IPS)
- VPN (Virtual Private Network)
- Secure communication protocols (HTTPS, SSL/TLS)

Importance:

Prevents external attackers from accessing the internal network.

3) Endpoint Security Layer

Explanation:

Focuses on securing end-user devices like computers, laptops, and mobile devices.

Measures:

- Antivirus and anti-malware software
- Device encryption
- Regular software updates

Importance:

Protects devices from malware and unauthorized access.

4) Application Security Layer

Explanation:

Protects software applications from vulnerabilities and attacks.

Measures:

- Secure coding practices
- Input validation
- Regular patching and updates

Importance:

Prevents attacks like SQL injection, XSS, and buffer overflow.

5) Data Security Layer

 Explanation:

Ensures protection of data at rest and in transit.

 Measures:

- Encryption techniques
- Data backup and recovery
- Access control mechanisms

 Importance:

Protects sensitive information from unauthorized access and loss.

6) Identity and Access Management (IAM) Layer

 Explanation:

Controls who can access what resources in the system.

 Measures:

- User authentication (passwords, biometrics)
- Authorization and role-based access
- Multi-factor authentication (MFA)

 Importance:

Ensures only authorized users can access resources.

7) Operational / Policy Security Layer

 Explanation:

Includes policies, procedures, and management practices for maintaining security.

 Measures:

- Security policies
- Monitoring and logging
- Incident response plans

 Importance:

Ensures proper handling of data and security incidents.

8) Human / User Awareness Layer

 Explanation:

Focuses on user behavior and awareness.

 Measures:

- Cyber security training
- Awareness of phishing attacks
- Strong password practices

 Importance:

Reduces risks caused by human errors.

Q4. Write a short note on : i) Cyber Stalking ii) Internet Governance in cyber security

ANS: i) Cyber Stalking

 Definition:

Cyber stalking is a form of cyber crime in which an individual uses the internet, social media, emails, or other digital communication tools to harass, threaten, monitor, or repeatedly contact a person without consent, causing fear or emotional distress.

Detailed Explanation:

- **Cyber stalking involves continuous and unwanted online behavior directed at a specific person.**
 - **The attacker may track the victim's online activities, send threatening messages, or misuse personal information.**
 - **It often starts as harmless communication but gradually becomes harassment.**
 - **It violates privacy, safety, and mental well-being of the victim.**
-

Techniques Used in Cyber Stalking:

- **Sending repeated emails, messages, or comments**
 - **Monitoring social media profiles and activities**
 - **Hacking accounts to gather personal data**
 - **Posting false or harmful content**
 - **Impersonation (fake profiles)**
 - **Tracking location using digital tools**
-

Effects of Cyber Stalking:

- **Mental stress, anxiety, and fear**
 - **Loss of privacy**
 - **Damage to reputation**
 - **Emotional trauma**
 - **In extreme cases, physical danger**
-

Prevention Measures:

- **Avoid sharing personal information online**

- Use strong passwords and privacy settings
 - Block and report suspicious accounts
 - Do not respond to unknown messages
 - Keep social media accounts private
-

Legal Aspects:

- Cyber stalking is punishable under cyber laws (like IT Act in India)
 - Victims can report incidents to cyber crime authorities
-

Conclusion:

Cyber stalking is a serious cyber threat that affects individuals emotionally and socially. Proper awareness, preventive measures, and legal actions are essential to control it.

ii) Internet Governance in Cyber Security

Definition:

Internet governance refers to the development and application of rules, policies, standards, and decision-making processes that regulate the use and operation of the internet, ensuring it remains secure, stable, and accessible.

Detailed Explanation:

- Internet governance involves collaboration between governments, private organizations, and international bodies.
- It ensures that the internet operates smoothly while maintaining security, privacy, and fairness.
- It plays a crucial role in managing cyber threats and enforcing cyber laws.

Objectives:

- Ensure secure and reliable internet
 - Protect user data and privacy
 - Prevent cyber crimes
 - Promote fair and equal access
 - Maintain global coordination
-

Key Components:

- Domain name system management
 - IP address allocation
 - Cyber laws and policies
 - Data protection regulations
 - Security standards and protocols
-

Organizations Involved:

- ICANN – manages domain names and IP addresses
 - International Telecommunication Union – sets global communication standards
 - National cyber security agencies (like CERT)
-

Importance in Cyber Security:

- Helps prevent cyber attacks and misuse of internet
- Ensures safe communication and data transfer
- Protects digital infrastructure
- Maintains trust in online systems

 Challenges:

- **Global coordination between countries**
- **Balancing privacy and security**
- **Handling rapidly increasing cyber threats**
- **Managing misuse of internet (fake news, cyber crime)**

Q5. Write short notes on i) Malware ii) Phishing iii) MITA attack iv) Malware Phishing Cyber threat

ANS:

i) Malware

 Definition:

Malware (Malicious Software) is a program or code designed to damage, disrupt, or gain unauthorized access to a computer system or network without the user's knowledge.

 Explanation:

- **Malware is created by attackers to harm systems or steal information.**
- **It can enter a system through email attachments, infected websites, USB drives, or downloads.**
- **Once installed, it may run silently in the background and affect system performance or security.**

 Types of Malware:

- **Virus – attaches to files and spreads**
- **Worm – spreads automatically over networks**

- Trojan Horse – looks harmless but is harmful
 - Ransomware – locks data and demands money
 - Spyware – secretly collects user information
-

Effects:

- Data theft or corruption
 - System slowdown or crash
 - Unauthorized access
 - Financial loss
-

Prevention:

- Install antivirus software
 - Avoid unknown downloads
 - Keep system updated
 - Use firewalls
-

ii) Phishing

Definition:

Phishing is a cyber attack in which attackers pretend to be a trusted entity to trick users into revealing sensitive information like passwords, OTPs, or bank details.

Explanation:

- Phishing is a form of social engineering attack.
- Attackers send fake emails, messages, or websites that look real.

- Users are tricked into entering personal information.
-

Types of Phishing:

- Email phishing
 - SMS phishing (smishing)
 - Voice phishing (vishing)
 - Fake websites
-

Effects:

- Identity theft
 - Financial fraud
 - Account hacking
-

Prevention:

- Check email sender carefully
 - Avoid clicking unknown links
 - Verify website URLs (HTTPS)
 - Never share OTP or passwords
-

iii) Man-in-the-Middle (MITM) Attack

Definition:

A Man-in-the-Middle (MITM) attack is a cyber attack where an attacker secretly intercepts communication between two parties and may alter the data without their knowledge.

Explanation:

- The attacker sits between sender and receiver.
 - Both think they are communicating directly.
 - Attacker can read, steal, or modify messages.
-

Effects:

- Password theft
 - Data interception
 - Financial loss
 - Privacy breach
-

Prevention:

- Use HTTPS websites
 - Use encryption (SSL/TLS)
 - Use VPN on public networks
 - Avoid unsecured Wi-Fi
-

iv) Cyber Threat (with Malware & Phishing)

Definition:

A cyber threat is any potential attack that can harm computer systems, steal data, or disrupt services.

Explanation:

- Cyber threats include all malicious activities performed on systems or networks.

- **Malware and phishing are two of the most common cyber threats.**
 - **These threats exploit vulnerabilities in systems or human behavior.**
-

Types of Cyber Threats:

- **Malware attacks**
 - **Phishing attacks**
 - **MITM attacks**
 - **DoS/DDoS attacks**
 - **SQL injection**
-

Effects:

- **Data loss**
 - **System damage**
 - **Financial fraud**
 - **Privacy violation**
-

Prevention:

- **Strong passwords**
- **Antivirus and firewall**
- **Security awareness**
- **Regular updates**

Q6. Elaborate different cyber security polycies in detail and explain different challenges in internet governance

ANS:

1) Cyber Security Policies

 **Definition:**

Cyber security policies are a set of rules, guidelines, and procedures designed to protect an organization's computers, networks, and data from cyber threats.

They define how security is maintained, who can access data, and how incidents are handled.

 **Types of Cyber Security Policies (Detailed)**

1) Access Control Policy

 **Explanation:**

This policy defines who can access which resources in a system.

 **Key Points:**

- **User authentication (passwords, OTP, biometrics)**
- **Role-based access control (admin, user, guest)**
- **Least privilege principle**

 **Purpose:**

Prevents unauthorized access to sensitive data.

2) Password Policy

 **Explanation:**

Defines rules for creating and managing secure passwords.

Key Points:

- Strong password (mix of letters, numbers, symbols)
- Regular password change
- No reuse of old passwords

Purpose:

Protects accounts from brute force and hacking attacks.

3) Data Protection Policy

Explanation:

Ensures that sensitive data is secure, confidential, and not misused.

Key Points:

- Data encryption
- Backup and recovery
- Controlled data sharing

Purpose:

Protects data from theft or loss.

4) Network Security Policy

Explanation:

Secures network infrastructure from unauthorized access and attacks.

Key Points:

- Firewalls and IDS/IPS
- VPN usage
- Secure protocols (HTTPS, SSL/TLS)

 Purpose:

Protects communication over networks.

5) Acceptable Use Policy (AUP)

 Explanation:

Defines proper use of organizational IT resources.

 Key Points:

- **No illegal downloads**
- **No misuse of internet**
- **Proper use of email and systems**

 Purpose:

Prevents misuse of resources.

6) Incident Response Policy

 Explanation:

Defines how to respond to cyber attacks or security breaches.

 Key Points:

- **Detection of attack**
- **Reporting mechanism**
- **Recovery steps**
- **Forensic analysis**

 Purpose:

Minimizes damage during cyber incidents.

7) Backup and Recovery Policy

Explanation:

Ensures regular backup of important data and recovery in case of loss.

Key Points:

- Regular backups
- Secure storage
- Disaster recovery plan

Purpose:

Prevents data loss due to attacks or system failure.

Q7. Explain the term cyber stalking. How to identify and detect cyber stalking.

ANS:

Cyber Stalking

Definition:

Cyber stalking is a cyber crime in which a person uses the internet, social media, emails, or other digital platforms to harass, threaten, monitor, or repeatedly contact another person without consent, causing fear, distress, or harm.

Explanation:

- Cyber stalking is a form of online harassment.
 - The attacker (stalker) repeatedly follows or targets a victim online.
 - It can include sending unwanted messages, tracking online activity, or spreading false information.
 - It affects the victim's privacy, safety, and mental health.
-

Characteristics:

- **Repeated unwanted communication**
 - **Use of fake profiles or anonymous accounts**
 - **Monitoring victim's online activity**
 - **Threatening or abusive messages**
 - **Emotional or psychological pressure**
-

Examples:

- **Repeated messages on social media**
 - **Fake accounts sending threats**
 - **Tracking someone's location through apps**
 - **Posting offensive content about victim**
-

How to Identify Cyber Stalking

Cyber stalking can be identified by observing unusual or suspicious online behavior.

1) Repeated Unwanted Contact:

- **Continuous messages, emails, or calls even after blocking**
 - **Persistent attempts to contact the victim**
-

2) Fake or Anonymous Profiles:

- **Unknown accounts interacting frequently**
 - **Multiple fake accounts targeting the same person**
-

3) Monitoring Online Activity:

- Someone commenting on every post or activity
 - Knowing personal details without being told
-

4) Threatening or Harassing Messages:

- Abusive language
 - Threats of harm or embarrassment
 - Emotional pressure messages
-

5) Unauthorized Tracking:

- Location tracking through apps or social media
 - Knowing private movements or habits
-

6) Sudden Privacy Breach:

- Personal photos or data shared without permission
 - Accounts being accessed or hacked
-

How to Detect Cyber Stalking

Detection involves technical and behavioral analysis.

1) Monitor Unusual Activity:

- Check login history of accounts
 - Look for unknown devices or locations
-

2) Use Security Tools:

- Enable notifications for login attempts
 - Use account activity logs on social media
-

3) Analyze Message Patterns:

- Frequent messages from unknown users
 - Repeated contact after blocking
-

4) Check IP Tracking (Advanced):

- IT security tools can trace repeated suspicious IP addresses
-

5) Report and Document Evidence:

- Save screenshots of messages
 - Report to cyber crime authorities
-

Prevention Measures:

- Do not share personal details publicly
 - Use strong privacy settings on social media
 - Block and report suspicious users
 - Enable two-factor authentication
 - Avoid accepting unknown friend requests
-

Legal Protection:

Cyber stalking is a punishable offense under cyber laws in many countries, including the IT Act in India. Victims can report it to cyber crime cells.

Q8. Explain the types of cyber-crimes.

ANS:

 **Definition:**

Cyber crimes are illegal activities carried out using computers, networks, or the internet to harm individuals, organizations, systems, or society.

 **1) Crimes Against Individuals**

- **Target: Single users or personal accounts**
- **These crimes directly affect a person's privacy, identity, or mental peace.**

 **Examples:**

- **Cyber stalking → repeated online harassment of a person**
- **Phishing → fake emails asking for bank password/OTP**
- **Identity theft → using someone's personal details for fraud**

 **Impact:**

- **Emotional stress**
 - **Privacy loss**
 - **Financial fraud**
-

 **2) Crimes Against Property**

- **Target: Data, software, or computer systems**
- **These crimes damage or steal digital resources.**

 **Examples:**

- **Virus attack deleting or corrupting files**
- **Ransomware locking system data and demanding money**

- **Software piracy (illegal copying of software)**

 Impact:

- **Data loss**
 - **System damage**
 - **Financial loss**
-

 3) Crimes Against Organizations

- **Target: Companies, banks, institutions**
- **These attacks affect business operations and reputation.**

 Examples:

- **DDoS attack on banking or company website**
- **Hacking organization database**
- **Corporate espionage (stealing business secrets)**

 Impact:

- **Financial loss**
 - **Service disruption**
 - **Reputation damage**
-

 4) Crimes Against Government

- **Target: Government systems and national infrastructure**
- **These are serious crimes affecting national security.**

 Examples:

- **Cyber terrorism (attacking government websites)**
- **Hacking defense or military systems**
- **Cyber warfare between countries**

 Impact:

- **National security threat**
 - **Political instability**
-

 5) Financial Cyber Crimes

- **Target: Money and financial systems**
- **These crimes involve online financial fraud.**

 Examples:

- **UPI fraud or fake payment links**
- **Credit card cloning**
- **Online banking hacking**

 Impact:

- **Direct financial loss**
 - **Banking insecurity**
-

 6) Social Media Cyber Crimes

- **Target: Social networking users**
- **These crimes happen through platforms like Instagram, Facebook, etc.**

 Examples:

- **Fake social media profiles**
- **Cyber bullying**
- **Spreading fake news or rumors**

 Impact:

- **Reputation damage**

- **Mental stress**
-

7) Crimes Against Society

- **Target: Large groups or public order**
- **These crimes affect social harmony and safety.**

Examples:

- **Child pornography**
- **Hate speech online**
- **Cyber terrorism content circulation**

Impact:

- **Social disturbance**
- **Psychological harm**

Q9. Write a short note on Software attacks & hardware attacks with example.

ANS:

1) Software Attacks

Definition:

Software attacks are malicious activities that target software systems such as operating systems, applications, and databases to steal data, disrupt services, or gain unauthorized access.

Detailed Explanation:

- **Software attacks are performed using malicious code or programs.**
- **They exploit bugs, vulnerabilities, or weak configurations in software.**
- **These attacks can be executed remotely through networks or internet.**

- They mainly affect the confidentiality, integrity, and availability of data.
-

Characteristics:

- Remote execution possible
 - No physical access needed
 - Uses malware or exploit code
 - Targets software vulnerabilities
-

Types of Software Attacks:

- Virus attack
 - Worm attack
 - Trojan horse
 - Ransomware
 - Spyware
 - SQL Injection
 - Phishing attacks
-

Examples:

- Virus: Infects files and corrupts data
 - Ransomware: Locks system files and demands money
 - SQL Injection: Attacks database using malicious queries
 - Trojan Horse: Hidden malicious program inside a useful software
-

Effects:

- Data theft or corruption

- **System crash or slowdown**
 - **Unauthorized access**
 - **Financial loss**
 - **Privacy violation**
-

Prevention:

- **Use updated antivirus software**
 - **Install security patches regularly**
 - **Avoid unknown downloads and links**
 - **Use firewalls and secure coding practices**
-

2) Hardware Attacks

Definition:

Hardware attacks are attacks that target the physical components of a computer system such as CPU, memory, storage devices, or network hardware to steal or manipulate data.

Detailed Explanation:

- **Hardware attacks require physical or close access to the system in many cases.**
 - **Attackers may install malicious devices or tamper with existing hardware.**
 - **These attacks are difficult to detect using normal software tools.**
 - **They can cause permanent damage or data leakage.**
-

Characteristics:

- **Physical interaction with system**
 - **Hard to detect remotely**
 - **Can bypass software security**
 - **May cause permanent damage**
-

Types of Hardware Attacks:

- **Hardware keylogger installation**
 - **Chip-level tampering**
 - **Side-channel attacks (power, timing analysis)**
 - **Device replacement or theft**
 - **Firmware attacks**
-

Examples:

- **Hardware Keylogger: Device installed between keyboard and CPU to capture keystrokes**
 - **Hard Disk Theft: Stealing storage device to access sensitive data**
 - **Chip Tampering: Modifying CPU or memory chips**
 - **Side-channel attack: Extracting secret data using power consumption patterns**
-

Effects:

- **Permanent data loss**
- **Unauthorized data access**
- **System failure**
- **Physical damage to system**

 Prevention:

- Physical security (locks, CCTV, access control)
- Secure server rooms
- Hardware authentication checks
- Regular inspection of devices

Feature	Software Attacks	Hardware Attacks
Target	Software systems	Physical components
Access	Remote possible	Physical access needed
Detection	Easier using tools	Difficult to detect
Tools used	Malware, exploits	Devices, tampering
Impact	Data/system damage	Permanent system damage

Q10. Explain the threats and vulnerabilities of the information security system.

ANS:

Threats and Vulnerabilities of Information Security System

 1) Introduction:

An Information Security System is designed to protect data, networks, and computer systems from unauthorized access, damage, or misuse. However, it faces threats and vulnerabilities that can compromise security.

 2) Threats in Information Security

 Definition:

A threat is any potential danger or event that can exploit a vulnerability and cause harm to a system or data.

Types of Threats:

1) Malware Threats

- **Malicious software that damages systems**
 - **Examples: virus, worm, ransomware, Trojan horse**
 - **Effect: data loss, system corruption**
-

2) Phishing Threats

- **Fake emails or websites to steal information**
 - **Example: fake bank login pages**
 - **Effect: password theft, financial fraud**
-

3) Denial of Service (DoS) Threats

- **Overloads system with traffic**
 - **Example: DDoS attack on websites**
 - **Effect: system becomes unavailable**
-

4) Man-in-the-Middle (MITM) Threats

- **Intercepts communication between two users**
 - **Example: hacking data in public Wi-Fi**
 - **Effect: data theft or modification**
-

5) Insider Threats

- Threats from employees or trusted users
 - Example: employee stealing company data
 - Effect: data leakage, misuse of access
-

6) Social Engineering Threats

- Manipulating users to reveal information
 - Example: pretending to be IT support
 - Effect: unauthorized access
-

3) Vulnerabilities in Information Security

Definition:

A vulnerability is a weakness or flaw in a system that can be exploited by a threat to cause damage.

Types of Vulnerabilities:

1) Software Vulnerabilities

- Bugs or coding errors in software
 - Example: SQL injection, buffer overflow
 - Effect: system hacking, data theft
-

2) Hardware Vulnerabilities

- Weakness in physical devices
- Example: hardware keylogger, chip tampering

- **Effect: data leakage, system damage**
-

3) Network Vulnerabilities

- **Weakness in network security**
 - **Example: open ports, weak Wi-Fi password**
 - **Effect: unauthorized access, MITM attack**
-

4) Configuration Vulnerabilities

- **Improper system settings**
 - **Example: default passwords, misconfigured firewall**
 - **Effect: easy system exploitation**
-

5) Human Vulnerabilities

- **Mistakes by users**
 - **Example: clicking phishing links, weak passwords**
 - **Effect: account compromise, data theft**
-

Relationship Between Threats and Vulnerabilities:

- **Threat = Danger (attacker or attack method)**
 - **Vulnerability = Weakness in system**
 - **Attack happens when threat exploits vulnerability**
-

Example:

- **Weak password (vulnerability) + hacker attack (threat) = account breach**

Q11. Define the term cyber security and Explain Layers of Cyber Security in detail.

ANS:

Cyber Security

 **Definition:**

Cyber security is the practice of protecting computer systems, networks, devices, and data from unauthorized access, cyber attacks, damage, or theft using technologies, policies, and procedures.

 **Explanation:**

Cyber security ensures that digital systems remain safe from various cyber threats such as hacking, malware, phishing, and data breaches. It focuses on maintaining the CIA Triad:

- **Confidentiality** – only authorized users can access data
 - **Integrity** – data should not be modified or tampered
 - **Availability** – systems and data should be available when needed
-

 **Objectives of Cyber Security:**

- **Protect sensitive information**
- **Prevent unauthorized access**
- **Ensure secure communication**
- **Detect and respond to cyber attacks**
- **Maintain system reliability**

Cyber Security

 **Definition:**

Cyber security is the practice of protecting computer systems, networks, devices, and data from unauthorized access, cyber attacks, damage, or theft using technologies, policies, and procedures.

Explanation:

Cyber security ensures that digital systems remain safe from various cyber threats such as hacking, malware, phishing, and data breaches. It focuses on maintaining the CIA Triad:

- Confidentiality – only authorized users can access data
 - Integrity – data should not be modified or tampered
 - Availability – systems and data should be available when needed
-

Objectives of Cyber Security:

- Protect sensitive information
- Prevent unauthorized access
- Ensure secure communication
- Detect and respond to cyber attacks
- Maintain system reliability

Q12. What is a man-in-the-middle attack (MIM)? Explain in detail.

ANS:

Man-in-the-Middle (MITM) Attack

Definition:

A Man-in-the-Middle (MITM) attack is a type of cyber attack in which an attacker secretly intercepts and possibly alters communication between two parties (sender and receiver) without their knowledge.

Explanation:

- In normal communication, data flows directly between sender and receiver.
 - In MITM attack, the attacker secretly places himself in between the communication path.
 - Both sender and receiver think they are communicating directly, but actually all data passes through the attacker.
 - The attacker can read, steal, modify, or inject fake information.
-

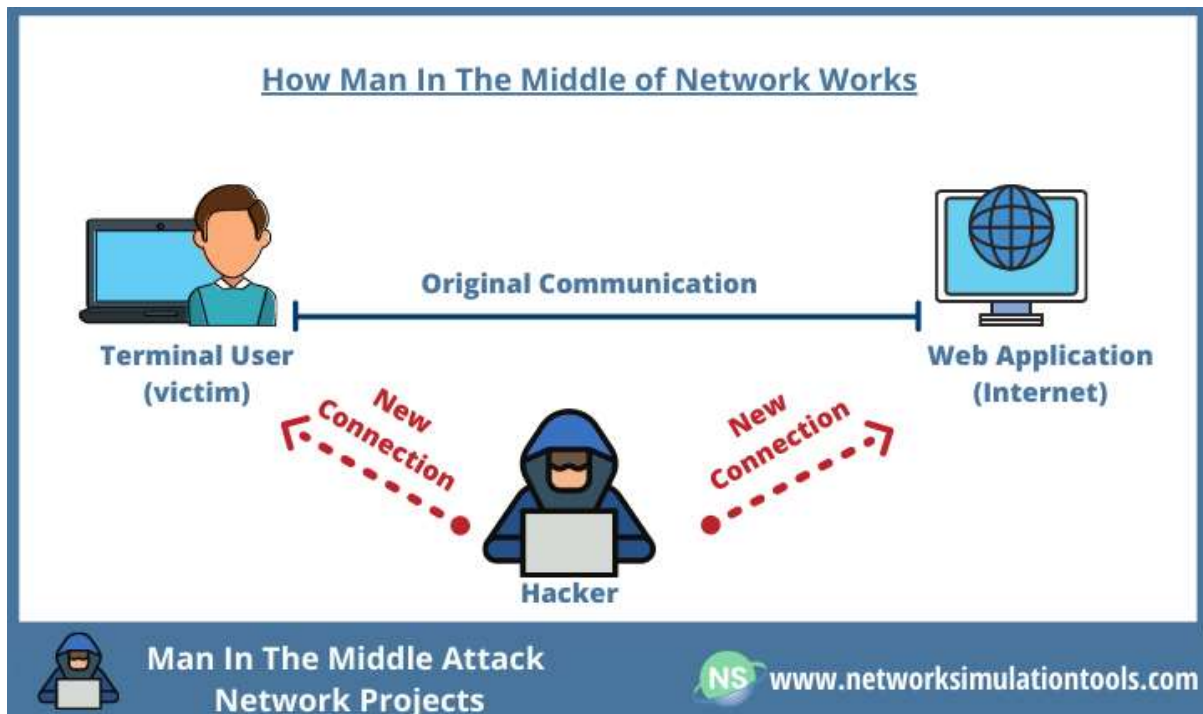
Working of MITM Attack:

1. A sender sends data to the receiver.
 2. Attacker intercepts the communication link.
 3. Attacker captures the data being transmitted.
 4. Attacker may modify the data or simply forward it.
 5. Receiver receives the message, unaware of interference.
-

Diagram:

Sender $\leftrightarrow$ Attacker $\leftrightarrow$ Receiver

(Intercepts communication)



Types of MITM Attacks:

- IP spoofing – attacker uses fake IP address
- DNS spoofing – redirects user to fake website
- Session hijacking – steals active session cookies
- Wi-Fi eavesdropping – attacks on public Wi-Fi networks
- Email interception – reading or modifying emails

Effects of MITM Attack:

- Theft of sensitive information (passwords, OTPs)
- Financial fraud
- Identity theft
- Data manipulation
- Loss of privacy

Example:

If a user logs into a banking website using public Wi-Fi, an attacker can intercept the login credentials and misuse them to access the bank account.

Prevention Methods:

- Use HTTPS (SSL/TLS encryption) websites
- Avoid using public Wi-Fi for sensitive transactions
- Use VPN (Virtual Private Network)
- Enable multi-factor authentication (MFA)
- Verify website authenticity before login

Q13. What is Cyber terrorism? What is an example of cyber terrorism?

ANS:

Cyber Terrorism

Definition:

Cyber terrorism is the deliberate use of computer systems, networks, and the internet by terrorist groups or individuals to carry out attacks that cause fear, disruption, damage, or destruction of critical infrastructure, government systems, or public services, with the intention of achieving political, ideological, or religious goals.

Detailed Explanation:

- Cyber terrorism is a combination of cyber space + terrorism activities.
- Unlike normal cyber crime (done for money or personal gain), cyber terrorism is done to create fear in society or destabilize a nation.

- **It targets critical infrastructure systems such as:**
 - **Government databases**
 - **Banking systems**
 - **Power grids**
 - **Communication networks**
 - **Defense systems**
 - **The main objective is not only data theft but also psychological impact on society, such as panic and insecurity.**
-

Key Characteristics of Cyber Terrorism:

- **Politically or ideologically motivated**
 - **Targets critical national infrastructure**
 - **Large-scale impact on society**
 - **Uses advanced hacking techniques and malware**
 - **Often coordinated by organized groups**
-

Methods Used in Cyber Terrorism:

- **Hacking government websites**
 - **Distributed Denial of Service (DDoS) attacks**
 - **Malware and ransomware attacks on critical systems**
 - **Data breaches of defense or financial systems**
 - **Spreading propaganda or fake information online**
 - **DNS spoofing and system manipulation**
-

Example of Cyber Terrorism:

- A terrorist group launching a DDoS attack on a country's banking system, causing online banking services to stop and creating financial panic among citizens.
 - Another example:
 - Hacking a government website and defacing it with terrorist messages or propaganda to spread fear and instability.
-

Effects of Cyber Terrorism:

- National security threats
 - Financial system disruption
 - Loss of public trust in digital systems
 - Economic damage
 - Psychological fear among citizens
 - Breakdown of essential services
-

Real-World Scenario (Example):

If attackers disable a country's power grid or hospital system using cyber attacks, it can lead to:

- Blackouts
- Medical emergencies
- Chaos in society

This is considered a form of cyber terrorism because it affects civilians and national stability.

Prevention Methods:

- Strong cyber security infrastructure

- **Firewalls, IDS, and IPS systems**
- **Encryption of sensitive data**
- **Continuous monitoring of critical systems**
- **Government cyber security agencies (CERT)**
- **Regular security audits and penetration testing**
- **Strict cyber laws and international cooperation**

Q14. Define Crime and Cybercrime. State Cybercrimes classification

ANS:

Crime and Cybercrime with Classification

1) Definition of Crime

Meaning:

A crime is an unlawful act or behavior that violates the laws of a country and is punishable by law.

Detailed Explanation:

- **Crime is any action that harms individuals, society, or government.**
 - **It is regulated by legal systems such as IPC (Indian Penal Code).**
 - **Crimes may be physical (theft, murder) or digital (cybercrime).**
 - **The main purpose of law is to prevent crime and maintain order in society.**
-

Examples:

- **Theft**

- **Fraud**
 - **Assault**
 - **Murder**
-

2) Definition of Cybercrime

Meaning:

A cybercrime is a crime in which computer systems, networks, or the internet are used as a tool, target, or medium to commit illegal activities.

Detailed Explanation:

- **Cybercrime is a modern form of crime that occurs in the digital environment.**
 - **It involves misuse of technology to steal data, disrupt systems, or harm users.**
 - **Cybercriminals use techniques like hacking, phishing, malware, and social engineering.**
 - **It affects individuals, organizations, and even governments.**
-

Examples:

- **Hacking bank accounts**
 - **Phishing emails**
 - **Identity theft**
 - **Ransomware attacks**
-

3) Cybercrime Classification

Cybercrimes are mainly classified based on the target of the crime:

A) Cybercrimes Against Individuals

Explanation:

These crimes directly target personal users and their data or privacy.

Details:

- Aim to steal personal information or harass individuals
- Commonly use social engineering techniques

Examples:

- Cyber stalking
- Phishing attacks
- Identity theft
- Email spoofing
- Online harassment

Impact:

- Mental stress
- Privacy violation
- Financial fraud

B) Cybercrimes Against Property

Explanation:

These crimes target digital property such as data, software, or systems.

Details:

- Focus on damaging or stealing digital assets
- Often involve malware or hacking tools

Examples:

- **Virus attacks**
- **Ransomware attacks**
- **Software piracy**
- **Data theft**
- **System hacking**

Impact:

- **Data loss**
 - **System corruption**
 - **Financial damage**
-

C) Cybercrimes Against Organizations

Explanation:

These crimes target businesses, companies, and institutions.

Details:

- **Aim to disrupt business operations or steal corporate data**
- **Often financially motivated or competitive sabotage**

Examples:

- **Distributed Denial of Service (DDoS) attacks**
- **Corporate data breaches**
- **Industrial espionage**
- **Website defacement**

Impact:

- **Business loss**

- Reputation damage
 - Service disruption
-

D) Cybercrimes Against Government

Explanation:

These are serious crimes targeting government systems and national infrastructure.

Details:

- Often politically or ideologically motivated
- Can threaten national security

Examples:

- Cyber terrorism
- Hacking government portals
- Cyber warfare attacks
- Stealing classified information

Impact:

- National security risk
 - Public panic
 - Government system disruption
-

E) Financial Cybercrimes

Explanation:

These crimes involve online financial fraud and theft of money.

Details:

- Targets banking systems and users' financial data

- **Highly common in digital transactions**

Examples:

- **UPI fraud**
- **Credit/debit card cloning**
- **Online banking hacking**
- **Fake payment links**

Impact:

- **Direct financial loss**
 - **Banking insecurity**
-

F) Cybercrimes Against Society

Explanation:

These crimes affect large groups of people or social order.

Details:

- **Aimed at disturbing peace and harmony**
- **Often spread through social media**

Examples:

- **Child pornography**
- **Hate speech**
- **Fake news spreading**
- **Cyber terrorism propaganda**

Impact:

- **Social unrest**
- **Psychological harm**

- **Public fear**

Q15. What Is Cyber stalking and cyber espionage explain with an example

How to identify and detect cyber stalking?

ANS:

Cyber Stalking

 **Definition:**

Cyber stalking is a cyber crime where a person uses the internet, social media, emails, or other digital platforms to harass, threaten, monitor, or repeatedly contact another person without consent, causing fear or emotional distress.

 **Explanation:**

- It is a form of online harassment.
 - The attacker repeatedly follows or targets the victim online.
 - It may include sending unwanted messages, fake accounts, or tracking online activity.
 - It affects the victim's privacy, safety, and mental health.
-

 **Example:**

- A person creates fake Instagram accounts and continuously sends threatening messages to another person even after being blocked.
-

 **Cyber Espionage**

 **Definition:**

Cyber espionage is the practice of using computers and networks to secretly access, steal, or monitor confidential information from individuals, organizations, or governments for strategic or political advantage.

Explanation:

- It is also called digital spying.
 - Attackers secretly collect sensitive data like government secrets, military plans, or corporate information.
 - It is usually carried out by hackers or nation-state actors.
-

Example:

- A hacker group breaking into a government database to steal defense information or secret military plans.
-

Difference in Simple Terms:

- Cyber stalking → targets individuals for harassment
 - Cyber espionage → targets organizations/government for secret data
-

How to Identify Cyber Stalking

Cyber stalking can be identified by observing unusual online behavior:

1) Repeated Unwanted Contact:

- Continuous messages, emails, or calls even after blocking
 - Persistent attempts to reach the victim
-

2) Fake or Anonymous Accounts:

- Unknown profiles repeatedly interacting
 - Multiple fake accounts targeting the same person
-

3) Monitoring Online Activity:

- Someone commenting on every post or activity
 - Knowing personal details without being told
-

4) Threatening Messages:

- Abusive or fear-creating messages
 - Blackmail or emotional pressure
-

5) Privacy Breach:

- Personal photos or information shared without permission
 - Unauthorized tracking of location
-

How to Detect Cyber Stalking

Detection involves both user awareness and technical monitoring:

1) Check Account Activity:

- Review login history on social media/email
 - Identify unknown devices or locations
-

2) Monitor Message Patterns:

- Repeated messages from unknown users
- Messages after blocking or ignoring

3) Use Security Tools:

- Privacy settings on social media
- Notifications for login attempts
- Anti-stalking or reporting features

4) IP and Network Tracking (Advanced):

- Security systems can trace repeated suspicious IP addresses

5) Evidence Collection:

- Save screenshots of messages
- Report to cyber crime authorities

Prevention Measures:

- Do not share personal details publicly
- Use strong privacy settings
- Block and report suspicious users
- Enable two-factor authentication
- Avoid accepting unknown friend requests

Q16. Explain the term Phishing & SQL Injection with a suitable example.

ANS:

Phishing

 **Definition:**

Phishing is a type of cyber attack in which attackers deceive users by pretending to be a trusted entity (like a bank, email service, or company) to steal sensitive information such as passwords, OTPs, credit card details, or personal data.

 **Detailed Explanation:**

- Phishing is mainly a social engineering attack, meaning it targets human psychology rather than technical systems.
 - Attackers create fake communication channels such as:
 - Fake emails
 - Fake SMS (smishing)
 - Fake websites
 - These messages create urgency like *“Your account will be blocked”* or *“Verify immediately”*.
 - The user is tricked into clicking a malicious link and entering confidential details.
 - Once entered, attackers capture and misuse the data.
-

 **Working Process:**

1. Attacker creates fake website/email
2. Sends message to victim
3. Victim clicks link
4. Victim enters credentials

5. Attacker steals data



Example:

A user receives an email:

"Your bank account has been suspended. Click here to verify."

The link opens a fake bank login page.

User enters username and password → attacker captures it → account is compromised.

Effects:

- Identity theft
- Financial fraud
- Account hacking
- Data leakage

Prevention:

- Check email sender carefully
- Avoid clicking unknown links
- Verify website URL (HTTPS)
- Never share OTP/password
- Enable multi-factor authentication

SQL Injection

Definition:

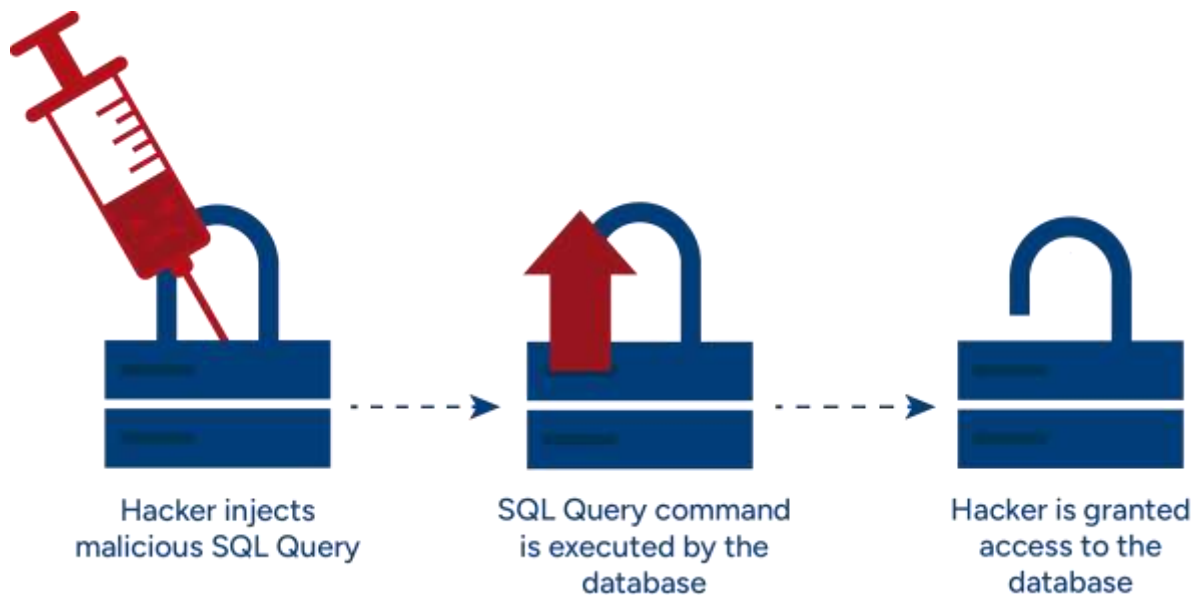
SQL Injection is a cyber attack in which an attacker injects malicious SQL commands into input fields of a web application to manipulate or gain unauthorized access to the database.

Detailed Explanation:

- Web applications interact with databases using SQL queries.
- If input validation is weak, attackers can insert SQL code into input fields.
- The database executes this malicious query along with normal commands.
- This allows attackers to:
 - View sensitive data
 - Modify records
 - Delete database tables
 - Bypass login authentication

Working Process:

1. Application accepts user input
2. Input is directly added to SQL query
3. Attacker inserts malicious SQL code
4. Database executes modified query
5. Unauthorized access occurs



Example:

Normal query:

```
SELECT * FROM users WHERE username = 'admin' AND password = '1234';
```

Attacker input:

username: admin' –

password: anything

Modified query becomes:

```
SELECT * FROM users WHERE username = 'admin' -- AND password = 'anything';
```

Explanation:

- `--` comments out the password check
 - Database only checks username
 - Attacker logs in without password
-

Effects:

- Unauthorized login
 - Data theft
 - Database corruption
 - Complete system compromise
-

Types of SQL Injection:

- Classic SQL injection
 - Blind SQL injection
 - Union-based injection
 - Error-based injection
-

Prevention:

- Use prepared statements (parameterized queries)
 - Validate user inputs strictly
 - Use stored procedures
 - Apply least privilege to database users
 - Use Web Application Firewall (WAF)
-

Key Difference (Understanding):

- Phishing → human-based attack (tricks users)
- SQL Injection → technical attack (exploits database vulnerability)

Q18. Explain various cyber crimes and Computer criminals

ANS:

Cyber Crimes and Computer Criminals

1) Cyber Crimes (Detailed Explanation)

Definition:

Cyber crimes are illegal activities carried out using computers, networks, or digital devices as a tool, target, or both, with the intent to harm individuals, organizations, or society.

Nature of Cyber Crimes:

- Occur in digital or online environment
 - Can be committed remotely from any location
 - Often anonymous and difficult to trace
 - May affect individuals, organizations, or governments
 - Can cause financial, social, or national damage
-

Classification of Cyber Crimes:

A) Crimes Against Individuals

Explanation:

These crimes directly target personal users and their private data.

Details:

- Aim to steal identity or harass individuals
- Use social engineering or hacking techniques

Examples:

- Cyber stalking (repeated harassment online)
- Phishing (fake emails/websites to steal passwords)
- Identity theft (using personal data illegally)
- Email spoofing (fake sender identity)

Impact:

- Emotional stress
 - Privacy breach
 - Financial fraud
-

B) Crimes Against Property

Explanation:

These crimes target digital assets like data, software, or systems.

Details:

- Focus on destruction or theft of digital resources
- Often involve malware or hacking tools

Examples:

- Virus attacks (file corruption)
- Worm attacks (self-spreading malware)
- Ransomware (data encryption for ransom)
- Software piracy (illegal copying of software)

- **Data theft**

 Impact:

- **Loss of important data**
 - **System failure**
 - **Financial damage**
-

 C) Crimes Against Organizations

 Explanation:

These crimes target companies, banks, and institutions.

 Details:

- **Aim to disrupt business operations**
- **May involve financial or competitive motives**

 Examples:

- **DDoS attacks (overloading servers)**
- **Corporate espionage (stealing trade secrets)**
- **Database hacking**
- **Website defacement**

 Impact:

- **Business loss**
 - **Reputation damage**
 - **Service downtime**
-

 D) Crimes Against Government

 Explanation:

These are serious crimes targeting national infrastructure and security systems.

 Details:

- Often politically or ideologically motivated
- Can be part of cyber warfare

 Examples:

- Cyber terrorism
- Hacking government portals
- Stealing defense information
- Disrupting public services

 Impact:

- National security threat
 - Public panic
 - Economic instability
-

 E) Financial Cyber Crimes

 Explanation:

These crimes involve fraud related to money and online transactions.

 Details:

- Target banking systems and users
- Highly increasing with digital payments

 Examples:

- UPI fraud
- Credit card cloning
- Online banking hacking

- Fake payment links

 Impact:

- Direct monetary loss
 - Loss of trust in digital banking
-

 F) Social Cyber Crimes

 Explanation:

These crimes occur on social media platforms.

 Details:

- Exploit user interaction platforms
- Often spread misinformation or abuse

 Examples:

- Fake profiles
- Cyber bullying
- Hate speech
- Fake news spreading

 Impact:

- Social unrest
 - Reputation damage
 - Psychological harm
-

 2) Computer Criminals (Cyber Criminals)

 Definition:

Computer criminals are individuals or groups who use computers and networks to commit illegal activities such as hacking, fraud, data theft, or disruption of services.

Characteristics of Computer Criminals:

- **Technically skilled or tool-based attackers**
 - **Operate anonymously using the internet**
 - **May act individually or in organized groups**
 - **Motivated by money, revenge, politics, or challenge**
-

Types of Computer Criminals:

1) Hackers

Explanation:

Hackers are individuals who access systems without permission.

Types:

- **White Hat Hackers → ethical hackers (security testing)**
- **Black Hat Hackers → malicious hackers (illegal access)**
- **Grey Hat Hackers → mix of ethical and unethical behavior**

Example:

Breaking into a company server without permission.

2) Crackers

Explanation:

Crackers are malicious hackers who break security systems illegally for damage or profit.

 **Example:**

Cracking software licenses to use paid software for free.

 **3) Phishers**

 **Explanation:**

Phishers create fake communication to steal sensitive data.

 **Example:**

Fake bank email asking for OTP or password.

 **4) Spammers**

 **Explanation:**

Spammers send unwanted bulk messages or emails, often containing ads or malware.

 **Example:**

Mass email advertising fake products or scams.

 **5) Cyber Stalkers**

 **Explanation:**

They harass or threaten individuals repeatedly online.

 **Example:**

Sending continuous threatening messages via social media.

6) Identity Thieves

Explanation:

They steal personal data to impersonate victims.

Example:

Using stolen Aadhaar or bank details for fraud.

7) Organized Cyber Criminal Groups

Explanation:

Highly skilled groups involved in large-scale cyber attacks.

Example:

Ransomware gangs attacking banks or hospitals.

Q19. What is the difference between Information security and cyber security?

ANS:

No.	Information Security	Cyber Security
1	Protects all types of information (digital + physical)	Protects only digital information and systems
2	Broader concept	Narrow concept
3	Includes protection of paper records, files, and verbal data	Focuses on computers, networks, and internet
4	Concerned with confidentiality, integrity, and availability of all information	Mainly focuses on online threats and cyber attacks
5	Can exist without computers	Depends on computer systems and networks
6	Protects data in any form (stored, printed, or spoken)	Protects data stored or transmitted electronically
7	Uses physical + digital security methods	Uses technical and network security tools
8	Example: Locking confidential documents in a cabinet	Example: Protecting a website from hacking
9	Used in organizations for overall data protection	Used to protect online systems and networks
10	Wider field including cyber security	Subset of information security